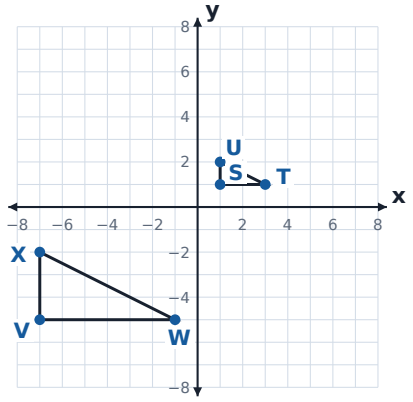
ANSWER KEY Reduced copy of the question page, with its answer

Is triangle STU similar to triangle VWX? If so, describe a similarity transformation; if not, explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q5/5

1601

QUESTION ID

S39-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. Dilate by 3 about S(1, 1), then translate 8 left and 6 down.

GRADING NOTE Any valid sequence. About S: T → (7, 1), U → (1, 4); then (x − 8, y − 6) gives V(−7, −5), W(−1, −5), X(−7, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 1

Similar figures as a sequence of transformations including a dilation • Grade 8 • Geometry • 8.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A figure and its image after a dilation and a translation are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q1/5

1603

QUESTION ID

S39-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A similarity sequence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q1/5

Two figures are similar. What may be different?

A. Their shape

B. Their size

C. Their angle measures

D. The ratios of their sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q2/5

1604

QUESTION ID

S39-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Their size

GRADING NOTE Size may change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle is dilated by 2 and then rotated. Its sides were 3, 4, 5. What are the image's sides?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q3/5

1605

QUESTION ID

S39-B1-Q3

TYPE

Short answer

CORRECT ANSWER

6, 8, 10

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q3/5

True or false?

Two figures with the same angles but different side ratios are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q4/5

1606

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Side ratios must all be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the transformation that changes size.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q5/5

1607

QUESTION ID

S39-B1-Q5

TYPE

Short answer

CORRECT ANSWER

A dilation

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 2

Specify a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Which rigid motion maps (x, y) to (−x, y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q1/5

1609

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Reflection across the y-axis

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which rigid motion maps (x, y) to (−x, −y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q2/5

1610

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Rotation of 180° about the origin

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q2/5

Which rigid motion maps (2, 3) to (−3, 2)?

A. Translation 5 left

B. Rotation 90° counterclockwise about the origin

C. Rotation 180°

D. Reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q3/5

1611

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Rotation 90° counterclockwise about the origin

GRADING NOTE (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q3/5

True or false?

A translation then a reflection is a valid sequence of rigid motions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q4/5

1612

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any composition of rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q4/5

True or false?

A dilation is a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q5/5

1613

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It changes lengths unless k = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 3

Dilate figures about the origin • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Dilate triangle (0, 0), (2, 0), (0, 1) by 3 about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q1/5

1615

QUESTION ID

S39-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(0, 0), (6, 0), (0, 3)

GRADING NOTE Form required: the three vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q1/5

True or false?

A dilation about the origin with factor 3 triples each coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q2/5

1616

QUESTION ID

S39-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (3x, 3y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q2/5

(4, −2) dilated by ½ about the origin is

A. (8, −4)

B. (2, −1)

C. (4.5, −1.5)

D. (2, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q3/5

1617

QUESTION ID

S39-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (2, −1)

GRADING NOTE Halve each coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A dilation about the origin maps (5, 5) to (2, 2). What is the factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q4/5

1618

QUESTION ID

S39-B3-Q4

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q4/5

True or false?

A dilation can change the angles of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q5/5

1619

QUESTION ID

S39-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angles are preserved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSG-SRT.A.2

FORWARD 1

Find the scale factor and correspondence when the image is turned or flipped • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has sides AB = 4, BC = 6, CA = 8. Triangle PQR has PQ = 9, QR = 12, RP = 6. Are they similar? If so, give the correspondence and the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q1/5

1621

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: A ↔ R, B ↔ P, C ↔ Q (AB ↔ RP = 6, BC ↔ PQ = 9, CA ↔ QR = 12); scale factor 1.5.

GRADING NOTE Form required: the correspondence and the factor. Match shortest to shortest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q1/5

A 3-4-5 triangle and a triangle with sides 7.5, 10, 12.5: similar? Scale factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q2/5

1622

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes; 2.5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q2/5

True or false?

Matching the shortest side with the shortest side, middle with middle, and longest with longest gives the correspondence for similar triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q3/5

1623

QUESTION ID
S39-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Similarity preserves the order of side lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q3/5

Triangles with sides 5, 6, 7 and 10, 12, 15: are they similar?

A. Cannot tell

B. Yes, factor 2.5

C. Yes, factor 2

D. No, 15/7 ≠ 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q4/5

1624

QUESTION ID

S39-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. No, 15/7 ≠ 2

GRADING NOTE Ratios 2, 2, 15/7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC ~ triangle DEF with a factor of 4 from ABC to DEF. DE = 10. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q5/5

1625

QUESTION ID

S39-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSG-SRT.A.2

FORWARD 2

Prove that all circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Circle C_1 has center A and radius r; circle C_2 has center B and radius s. Prove that C_1 and C_2 are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q1/5

1627

QUESTION ID

S39-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation and a dilation map C_1 onto C_2.

GRADING NOTE REQUIRED WORK: Translate A to B, then dilate about B with scale factor s/r; show the image of C_1 is C_2.

Any valid proof (paragraph, two-column, or flow). Essential: the translation maps C_1 to the circle with center B and radius r; the dilation about B with factor s/r sends each point at distance r from B to a point at distance s, so the image is C_2 (and every point of C_2 is reached).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle P has radius 4 and circle Q has radius 10. What scale factor maps P onto Q (after a translation)?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q2/5

1628

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q2/5

True or false?

Any two rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q3/5

1629

QUESTION ID

S39-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 1 × 2 and a 1 × 3 rectangle do not have proportional sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q3/5

Which pair of figures is NOT always similar?

A. Two equilateral triangles

B. Two squares

C. Two circles

D. Two ellipses

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q4/5

1630

QUESTION ID

S39-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Two ellipses

GRADING NOTE Ellipses can have different shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q4/5

Explain why all equilateral triangles are similar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q5/5

1631

QUESTION ID

S39-F2-Q5

TYPE

Short answer

CORRECT ANSWER

All their angles are 60°, so AA applies (or all side ratios are equal).

GRADING NOTE Any valid explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q5/5

Explain similarity of triangles as equal angles and proportional sides

MAIN	Geometry main question	**HSG-SRT.A.2**
BACKWARD 1	Read correspondence from a similarity or congruence statement	**HSG-CO.B.7**
BACKWARD 2	Solve proportions	**7.RP.A.2**
BACKWARD 3	Scale factor between figures	**7.G.A.1**
FORWARD 1	Use the AA criterion to recognize similar triangles	**HSG-SRT.A.3**
FORWARD 2	Use similarity to find unknown lengths in problems	**HSG-SRT.B.5**

SET 40 • HSG-SRT.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A similarity transformation (a dilation with scale factor k followed by rigid motions) maps △ABC onto △DEF with A → D, B → E, C → F. Explain why ∠A ≅ ∠D, ∠B ≅ ∠E, ∠C ≅ ∠F and DE/AB = EF/BC = FD/CA = k.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q1/5

1634

QUESTION ID

S40-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angles are preserved and every length is multiplied by k.

GRADING NOTE REQUIRED WORK: Use what dilations and rigid motions do to angles and to lengths.

Any valid explanation. Essential: dilations and rigid motions preserve angle measure, so corresponding angles are congruent; the dilation multiplies every length by k and rigid motions keep lengths, so each image side is k times its original side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q1/5

△ABC ~ △DEF with A ↔ D, B ↔ E, C ↔ F. AB = 6, BC = 9, DE = 10. Find EF and the scale factor from ABC to DEF.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q2/5

1635

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

EF = 15; scale factor 5/3

GRADING NOTE Form required: both. 10/6 = 5/3; EF = 9 · 5/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△PQR ~ △STU with P ↔ S, Q ↔ T, R ↔ U. m∠P = 42° and m∠Q = 71°. Find m∠U.

m∠U = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q3/5

1636

QUESTION ID

S40-M-Q3

TYPE

Short answer

CORRECT ANSWER

67°

GRADING NOTE Any equivalent form. ∠U ≅ ∠R = 180 − 113.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q3/5

△ABC has AB = 4, BC = 6, CA = 8, and △XYZ has XY = 6, YZ = 9, ZX = 10. Using the meaning of similarity, decide whether △ABC ~ △XYZ with A ↔ X, B ↔ Y, C ↔ Z, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q4/5

1637

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. XY/AB = 1.5 and YZ/BC = 1.5 but ZX/CA = 1.25; similar triangles need all three ratios equal (one scale factor).

GRADING NOTE Form required: all three ratios and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q4/5

A dilation with scale factor 2.5 followed by a rotation maps △JKL onto △MNO (J → M, K → N, L → O). m∠J = 50°, m∠K = 60°, JK = 4, KL = 6. Find m∠O, MN and NO, and explain each answer using what the transformations do.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q5/5

1638

QUESTION ID

S40-M-Q5

TYPE

Short answer

CORRECT ANSWER

m∠O = m∠L = 70° (angles are preserved); MN = 2.5 · 4 = 10 and NO = 2.5 · 6 = 15 (the dilation multiplies lengths by 2.5; the rotation keeps them).

GRADING NOTE Form required: three values, each with its reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 1

Read correspondence from a similarity or congruence statement • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

ANSWER KEY Reduced copy of the question page, with its answer

△ABC ~ △XYZ. Which angle corresponds to ∠C?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q1/5

1640

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

∠Z

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

△ABC ~ △XYZ. Which side corresponds to BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q2/5

1641

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

YZ

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△KLM ~ △RST. Which ratio equals KL/RS?

A. LM/RT

B. RS/LM

C. LM/ST

D. KM/ST

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q3/5

1642

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. LM/ST

GRADING NOTE Second pair of letters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\triangle DEF \sim \triangle GHI$ means $\angle E \cong \angle H$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q4/5

1643

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Middle letters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q4/5

True or false?

△DEF ~ △GHI means DF corresponds to GH.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q5/5

1644

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE DF ↔ GI.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 6/9 = 10/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q1/5

1646

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve x/4 = 15/6.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q2/5

1647

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which ratio is equal to 3 : 5?

A. 9 : 12

B. 6 : 8

C. 12 : 20

D. 5 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q3/5

1648

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12 : 20

GRADING NOTE Multiply both by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

2/3 = 8/12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q4/5

1649

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both reduce to 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

4/6 = 6/9 = 10/14

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q5/5

1650

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10/14 = 5/7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 3

Scale factor between figures • Grade 7 • Geometry • 7.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A figure is enlarged so that a 4 cm side becomes 10 cm. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q1/5

1652

QUESTION ID

S40-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q1/5

True or false?

All lengths of a scale drawing are multiplied by the same number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q2/5

1653

QUESTION ID

S40-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The scale factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q2/5

With scale factor 3, a 7 cm side becomes

A. 7 cm

B. 10 cm

C. 73 cm

D. 21 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q3/5

1654

QUESTION ID

S40-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 21 cm

GRADING NOTE Multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A scale drawing's 12 cm side represents 60 cm. Scale factor (drawing to actual)?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q4/5

1655

QUESTION ID

S40-B3-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{5}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A scale drawing changes the angles of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q5/5

1656

QUESTION ID

S40-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angles stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSG-SRT.A.2

FORWARD 1

Use the AA criterion to recognize similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles have angles of 35° and 80°. Are they similar? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q1/5

1658

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes, by AA: two pairs of congruent angles (the third angles are both 65°).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has m∠A = 50°, m∠B = 60°. Triangle DEF has m∠D = 60°, m∠F = 70°. Are they similar? Give the correspondence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q2/5

1659

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: ∠C = 70°, ∠E = 50°; A ↔ E, B ↔ D, C ↔ F.

GRADING NOTE Form required: the decision and the correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q2/5

True or false?

Two right triangles with one pair of congruent acute angles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q3/5

1660

QUESTION ID

S40-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two pairs of congruent angles (the right angles and the acute angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q3/5

Which pair is NOT necessarily similar?

A. Two triangles with angles 30° and 60°

B. Two right isosceles triangles

C. Two equilateral triangles

D. Two isosceles triangles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q4/5

1661

QUESTION ID

S40-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Two isosceles triangles

GRADING NOTE Isosceles triangles can have different vertex angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q4/5

In a figure, DE ∥ BC with D on AB and E on AC. Explain why △ADE ~ △ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q5/5

1662

QUESTION ID

S40-F1-Q5

TYPE

Short answer

CORRECT ANSWER

∠A is shared and ∠ADE ≅ ∠ABC (corresponding angles), so AA.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

FORWARD 2

Use similarity to find unknown lengths in problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A 1.8 m person casts a 2.4 m shadow when a tree casts a 16 m shadow. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q1/5

1664

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12 m

GRADING NOTE Any equivalent form. 1.8/2.4 = h/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC.
AD = 4, DB = 6, and DE = 5. Find BC.

BC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q2/5

1665

QUESTION ID
S40-F2-Q2

TYPE
Short answer

CORRECT ANSWER
12.5

GRADING NOTE Any equivalent form. AD/AB = DE/BC: 4/10 = 5/BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q2/5

True or false?

In similar triangles with scale factor 3, a 7 cm side corresponds to a 10 cm side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q3/5

1666

QUESTION ID

S40-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7 × 3 = 21 cm; a scale factor multiplies, it does not add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 6 ft pole casts a 4 ft shadow. At the same time, a building casts a 30 ft shadow. How tall is the building?

A. 36 ft

B. 45 ft

C. 20 ft

D. 24 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q4/5

1667

QUESTION ID

S40-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 45 ft

GRADING NOTE 6/4 = h/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two similar triangles have perimeters 24 and 36. A side of the smaller is 10. Find the corresponding side of the larger.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q5/5

1668

QUESTION ID

S40-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. Factor 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q5/5

Establish the AA criterion with similarity transformations

MAIN	Geometry main question	**HSG-SRT.A.3**
BACKWARD 1	Prove triangles congruent with ASA	**HSG-CO.B.8**
BACKWARD 2	Dilations multiply lengths and keep angles	**HSG-SRT.A.1.b**
BACKWARD 3	Use the angle sum to find a third angle	**8.G.A.5**
FORWARD 1	Establish the SAS and SSS similarity criteria	**HSG-SRT.A.3**
FORWARD 2	Use AA to set up and solve problems	**HSG-SRT.B.5**

SET 41 • HSG-SRT.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

In △ABC and △DEF, ∠A ≅ ∠D and ∠B ≅ ∠E. Prove that △ABC ~ △DEF using similarity transformations.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q1/5

1671

QUESTION ID

S41-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ~ △DEF.

GRADING NOTE REQUIRED WORK: Dilate △ABC by k = DE/AB; show the image is congruent to △DEF by ASA; conclude a dilation followed by rigid motions maps ABC onto DEF.

Any valid proof (paragraph, two-column, or flow). Essential: (1) dilate by k = DE/AB to get △A′B′C′ with A′B′ = DE; (2) dilations keep angles, so ∠A′ ≅ ∠D and ∠B′ ≅ ∠E; (3) △A′B′C′ ≅ △DEF by ASA; (4) so rigid motions map A′B′C′ onto DEF, and the dilation followed by those motions maps ABC onto DEF. Invalid: assuming the sides are proportional.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR and △STU, m∠P = m∠S = 40°, m∠Q = m∠T = 75°, PQ = 4 and ST = 10. Prove that △PQR ~ △STU by describing a similarity transformation and justifying each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q2/5

1672

QUESTION ID

S41-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

△PQR ~ △STU (scale factor 2.5)

GRADING NOTE REQUIRED WORK: Dilate △PQR by 10/4 = 2.5, show the image is congruent to △STU by ASA, and conclude.

Any valid proof (paragraph, two-column, or flow). Essential: dilation with factor 2.5 gives △P′Q′R′ with P′Q′ = 10 = ST; angles are preserved, so ∠P′ ≅ ∠S and ∠Q′ ≅ ∠T; ASA gives △P′Q′R′ ≅ △STU; the dilation followed by the rigid motions maps PQR onto STU.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q2/5

Prove that two right triangles with one pair of congruent acute angles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q3/5

1673

QUESTION ID

S41-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The triangles are similar by AA.

GRADING NOTE

REQUIRED WORK: Name the triangles, list the two pairs of congruent angles, and apply AA (stating why AA is valid).

Any valid proof (paragraph, two-column, or flow). Essential: the right angles are congruent; the given acute angles are congruent; two pairs of congruent angles; AA. Invalid: assuming the sides are proportional.

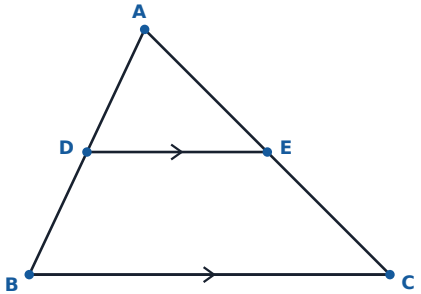
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that △ADE ~ △ABC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q4/5

1674

QUESTION ID

S41-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

△ADE ~ △ABC

GRADING NOTE REQUIRED WORK: Find two pairs of congruent angles using the parallel lines, then apply AA.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A ≅ ∠A (shared); ∠ADE ≅ ∠ABC (corresponding angles, DE ∥ BC, transversal AB); AA. Invalid: using side ratios not given.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that two isosceles triangles with congruent vertex angles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q5/5

1675

QUESTION ID

S41-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The triangles are similar by AA.

GRADING NOTE REQUIRED WORK: Let the vertex angles both measure v; find each base angle; apply AA.

Any valid proof (paragraph, two-column, or flow). Essential: base angles of an isosceles triangle are congruent, so each base angle measures (180° − v)/2 in both triangles; the vertex angles are congruent; two pairs of congruent angles; AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 1

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles have ∠A ≅ ∠D, AB = DE, ∠B ≅ ∠E. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q1/5

1677

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

ASA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q1/5

True or false?

ASA needs the side between the two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q2/5

1678

QUESTION ID

S41-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q2/5

Which set of parts gives ASA?

A. Three angles

B. Three sides

C. Two sides and the angle between them

D. Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q3/5

1679

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Two angles and the side between them

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q3/5

After proving △PQR ≅ △XYZ, are their corresponding sides equal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q4/5

1680

QUESTION ID

S41-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Corresponding parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q4/5

True or false?

ASA works only for right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q5/5

1681

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 2

Dilations multiply lengths and keep angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

A 5 cm segment is dilated by 1.6. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q1/5

1683

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-B2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q1/5

True or false?

A dilation preserves angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q2/5

1684

QUESTION ID

S41-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similar shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A dilation maps AB = 4 to A′B′ = 10. Factor?

A. 2.5

B. 6

C. 0.4

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q3/5

1685

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.5

GRADING NOTE 10 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What scale factor makes a 6 cm segment 6 cm long?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q4/5

1686

QUESTION ID

S41-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q4/5

True or false?

A dilation by 2 doubles the angles of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q5/5

1687

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angles are unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 3

Use the angle sum to find a third angle • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles 47° and 88°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q1/5

1689

QUESTION ID

S41-B3-Q1

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q1/5

True or false?

If two angles of one triangle equal two angles of another, the third angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q2/5

1690

QUESTION ID

S41-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Angle sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q2/5

Triangles with angles 30°, 70°, x and 70°, 80°, y. Which is true?

A. x = 80, y = 30; the triangles are similar

B. x = 80, y = 30; not similar

C. x = 70, y = 30

D. x = 30, y = 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q3/5

1691

QUESTION ID

S41-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 80, y = 30; the triangles are similar

GRADING NOTE 180 − 100 = 80; 180 − 150 = 30; same angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle are equal and the third is 40°. Find the equal angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q4/5

1692

QUESTION ID

S41-B3-Q4

TYPE

Short answer

CORRECT ANSWER

70° each

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q4/5

True or false?

Knowing one angle of each triangle is enough for similarity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q5/5

1693

QUESTION ID

S41-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Need two pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSG-SRT.A.3

FORWARD 1

Establish the SAS and SSS similarity criteria • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC and △DEF, DE/AB = DF/AC = k and ∠A ≅ ∠D. Prove that △ABC ~ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q1/5

1695

QUESTION ID

S41-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ~ △DEF (SAS similarity).

GRADING NOTE REQUIRED WORK: Dilate △ABC by k, then use SAS congruence with △DEF.

Any valid proof (paragraph, two-column, or flow). Essential: dilate by k: A′B′ = DE, A′C′ = DF, ∠A′ ≅ ∠A ≅ ∠D; SAS congruence; dilation then rigid motions map ABC onto DEF.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q1/5

Triangles with sides 4, 6, 8 and 6, 9, 12. Similar? Which criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q2/5

1696

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, SSS similarity (ratio 1.5).

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q2/5

True or false?

Two triangles with two pairs of proportional sides and congruent included angles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q3/5

1697

QUESTION ID

S41-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE SAS similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q3/5

Which is NOT a similarity criterion?

A. SAS

B. SSA

C. SSS

D. AA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q4/5

1698

QUESTION ID

S41-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. SSA

GRADING NOTE SSA does not determine shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

△PQR: PQ = 5, PR = 7, ∠P = 40°. △XYZ: XY = 10, XZ = 14, ∠X = 40°. Similar? Factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q5/5

1699

QUESTION ID

S41-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes (SAS similarity); factor 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSG-SRT.A.3

FORWARD 2

Use AA to set up and solve problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Two vertical poles are 3 m and 7 m tall. A wire from the top of each to the bottom of the other crosses at a point P. How high above the ground is P? (Use similar triangles.)

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q1/5

1701

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2.1 m

GRADING NOTE Any equivalent form. h = (3 · 7)/(3 + 7) = 2.1, from two pairs of AA-similar triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with ∠ADE ≅ ∠ACB. AD = 4, AE = 6, AB = 12. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q2/5

1702

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. △ADE ~ △ACB (AA): AD/AC = AE/AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q2/5

True or false?

A mirror on the ground can be used to find a building's height with similar triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q3/5

1703

QUESTION ID

S41-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal angles of incidence and reflection give AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q3/5

Diagonals of trapezoid ABCD (AB ∥ DC) meet at E. AB = 12, DC = 4, AE = 9. Find EC.

A. 6

B. 27

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q4/5

1704

QUESTION ID

S41-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE △ABE ~ △CDE (AA): EC = 9 · 4/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 2 m stick casts a 1.5 m shadow. A flagpole casts a 9 m shadow. How tall is the flagpole?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q5/5

1705

QUESTION ID
S41-F2-Q5

TYPE
Short answer

CORRECT ANSWER
12 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q5/5

Prove the triangle proportionality theorem

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Use AA to show triangles are similar	**HSG-SRT.A.3**
BACKWARD 2	Corresponding angles of parallel lines	**HSG-CO.C.9**
BACKWARD 3	Rewrite and solve proportions algebraically	**7.RP.A.2**
FORWARD 1	Prove that parallel lines cut transversals proportionally	**HSG-SRT.B.4**
FORWARD 2	Prove the angle bisector theorem	**HSG-SRT.B.5**

SET 42 • HSG-SRT.B.4

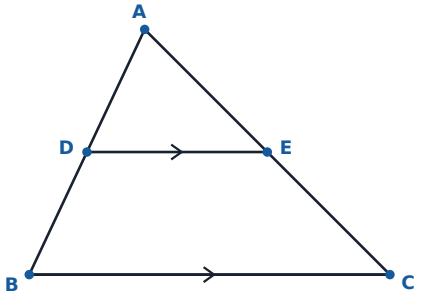
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/DB = AE/EC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q1/5

1708

QUESTION ID

S42-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AD/DB = AE/EC

GRADING NOTE REQUIRED WORK: Prove △ADE ~ △ABC by AA, write AD/AB = AE/AC, then rewrite with AB = AD + DB and AC = AE + EC.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A shared; ∠ADE ≅ ∠ABC (corresponding angles, DE ∥ BC); AA; AD/AB = AE/AC; so AB/AD = AC/AE, (AD + DB)/AD = (AE + EC)/AE, 1 + DB/AD = 1 + EC/AE, DB/AD = EC/AE, and AD/DB = AE/EC.

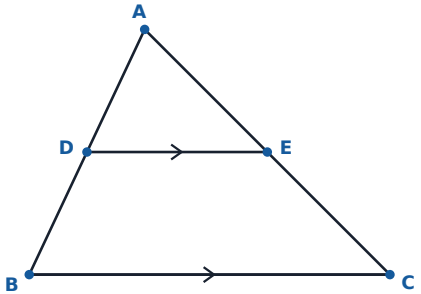
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/AB = AE/AC = DE/BC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q2/5

1709

QUESTION ID

S42-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

AD/AB = AE/AC = DE/BC

GRADING NOTE REQUIRED WORK: Prove △ADE ~ △ABC by AA, then use that corresponding sides of similar triangles are proportional.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠A; ∠ADE ≅ ∠ABC (corresponding angles); AA; corresponding sides proportional with AD ↔ AB, AE ↔ AC, DE ↔ BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR, a line parallel to QR meets PQ at S and PR at T. Prove that PS/SQ = PT/TR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q3/5

1710

QUESTION ID

S42-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

PS/SQ = PT/TR

GRADING NOTE REQUIRED WORK: AA similarity of △PST and △PQR, then the same algebra.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠P and corresponding angles give △PST ~ △PQR; then the proportion algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Coordinate proof. Place $\triangle ABC$ with $A(0, 0)$, $B(p, 0)$ and $C(q, r)$, where $p > 0$ and $r > 0$. $D(tp, 0)$ is on AB with $0 < t < 1$, and E is on AC with $DE \parallel BC$. Prove that $AD/DB = AE/EC$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q4/5

1711

QUESTION ID

S42-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AD/DB = AE/EC = t/(1 − t)

GRADING NOTE REQUIRED WORK: Write $E = (sq, sr)$ with $0 < s < 1$, use equal slopes of DE and BC to show $s = t$, then compute both ratios.

Any valid proof (paragraph, two-column, or flow). Essential: slope $DE = sr/(sq - tp)$ and slope $BC = r/(q - p)$ (or both vertical); equal slopes give $s(q - p) = sq - tp$, so $sp = tp$ and $s = t$; $AD = tp$, $DB = (1 - t)p$; $AE = t \cdot AC$, $EC = (1 - t) \cdot AC$; both ratios are $t/(1 - t)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/DB = AE/EC using the dilation about A with scale factor k = AB/AD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q5/5

1712

QUESTION ID

S42-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

AD/DB = AE/EC

GRADING NOTE REQUIRED WORK: Show the dilation maps D to B and line DE to line BC, so it maps E to C; compare lengths, then rewrite the proportion.

Any valid proof (paragraph, two-column, or flow). Essential: the dilation sends D to B (B is on ray AD with AB = k · AD); it sends line DE to the line through B parallel to DE, which is line BC; E is on ray AC, so its image is on ray AC and on line BC, which is C; hence AC = k · AE, so AD/AB = AE/AC = 1/k; then (AB − AD)/AD = (AC − AE)/AE gives DB/AD = EC/AE and AD/DB = AE/EC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 1

Use AA to show triangles are similar • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two pairs of congruent angles make two triangles similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q1/5

1714

QUESTION ID

S42-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Name two pairs of congruent angles in △ADE and △ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q2/5

1715

QUESTION ID
S42-B1-Q2

TYPE
Short answer

CORRECT ANSWER
∠A ≅ ∠A and ∠ADE ≅ ∠ABC (or ∠AED ≅ ∠ACB)

GRADING NOTE Any two valid pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q2/5

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Which ratio equals AD/AB?

A. DB/AB

B. AE/AC

C. DE/AB

D. AE/EC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q3/5

1716

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. AE/AC

GRADING NOTE Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. AD = 3, AB = 9, AE = 4. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q4/5

1717

QUESTION ID
S42-B1-Q4

TYPE
Short answer

CORRECT ANSWER
12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q4/5

True or false?

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Then AD/DB = DE/BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q5/5

1718

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE DB is not a side of either triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 2

Corresponding angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

When a transversal crosses parallel lines, corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q1/5

1720

QUESTION ID

S42-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC, so AB is a transversal of DE and BC. Which angle corresponds to ∠ADE?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q2/5

1721

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

∠ABC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q2/5

In △ABC, D is on AB and E is on AC with DE ∥ BC. With AC as the transversal, ∠AED corresponds to

A. ∠BAC

B. ∠ABC

C. ∠DEC

D. ∠ACB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q3/5

1722

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ∠ACB

GRADING NOTE Same position.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. m∠ABC = 65°. Find m∠ADE.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q4/5

1723

QUESTION ID
S42-B2-Q4

TYPE
Short answer

CORRECT ANSWER
65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Corresponding angles are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q5/5

1724

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Congruent for parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 3

Rewrite and solve proportions algebraically • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

If a/b = c/d, where a, b, c and d are all nonzero, write the equation you get by taking reciprocals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q1/5

1726

QUESTION ID

S42-B3-Q1

TYPE

Short answer

CORRECT ANSWER

b/a = d/c

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 4/(4 + x) = 6/15.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q2/5

1727

QUESTION ID

S42-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 60 = 24 + 6x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If (a + b)/a = (c + d)/c, then

A. b/a = d/c

B. b = d

C. a = c

D. a/b = c/d is false

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q3/5

1728

QUESTION ID

S42-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. b/a = d/c

GRADING NOTE Subtract 1 from both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

3/5 = 6/10 implies 5/3 = 10/6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q4/5

1729

QUESTION ID

S42-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q4/5

True or false?

(2 + 3)/2 = 5/2 implies 3/2 = 5/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q5/5

1730

QUESTION ID

S42-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtracting 1 gives 3/2 = 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • HSG-SRT.B.4

FORWARD 1

Prove that parallel lines cut transversals proportionally • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Three parallel lines cut transversal t at A, B, C and transversal u at D, E, F (in order). Prove that AB/BC = DE/EF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q1/5

1732

QUESTION ID

S42-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB/BC = DE/EF

GRADING NOTE

REQUIRED WORK: Draw segment AF; apply the triangle proportionality theorem in △ACF and in △FDA.

Any valid proof (paragraph, two-column, or flow). Essential: AF crosses the middle line at G; in △ACF, BG ∥ CF so AB/BC = AG/GF; in △FAD, GE ∥ AD so FG/GA = FE/ED, so AG/GF = DE/EF; combine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three parallel lines cut one transversal into segments 4 and 6 and another into segments 5 and x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q2/5

1733

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q2/5

True or false?

Parallel lines that cut equal segments on one transversal cut equal segments on every transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q3/5

1734

QUESTION ID

S42-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Ratio 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 5, DB = 3, AE = 7.5. Find EC.

A. 12.5

B. 4.5

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q4/5

1735

QUESTION ID

S42-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4.5

GRADING NOTE 5/3 = 7.5/EC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Lots between parallel streets have frontages 30 m and 45 m on one avenue. On another avenue the first frontage is 40 m. Find the second.

Frontage: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q5/5

1736

QUESTION ID
S42-F1-Q5

TYPE
Short answer

CORRECT ANSWER
60 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

FORWARD 2

Prove the angle bisector theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, the bisector of ∠A meets BC at D. Prove that BD/DC = AB/AC. (Draw the line through C parallel to AD, meeting line AB at E.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q1/5

1738

QUESTION ID

S42-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

BD/DC = AB/AC

GRADING NOTE REQUIRED WORK: Show △ACE is isosceles (AE = AC), then apply the triangle proportionality theorem in △BCE.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAD ≅ ∠AEC (corresponding) and ∠DAC ≅ ∠ACE (alternate interior), and ∠BAD ≅ ∠DAC, so ∠AEC ≅ ∠ACE and AE = AC; in △BCE, AD ∥ EC gives BD/DC = BA/AE = AB/AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, the bisector of ∠A meets BC at D. AB = 9, AC = 6, BC = 10. Find BD and DC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q2/5

1739

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

BD = 6, DC = 4

GRADING NOTE Form required: both. BD/DC = 9/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q2/5

True or false?

The bisector of an angle of a triangle always meets the opposite side at its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q3/5

1740

QUESTION ID

S42-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when the two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The bisector of ∠P in △PQR divides QR into 3 and 5. PQ = 6. Find PR.

A. 10

B. 15

C. 3.6

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q4/5

1741

QUESTION ID

S42-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE 3/5 = 6/PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the angle bisector theorem is a relationship that similarity or proportionality, not congruence, proves.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q5/5

1742

QUESTION ID
S42-F2-Q5

TYPE
Short answer

CORRECT ANSWER
It states a ratio of lengths, which comes from proportional segments (parallel lines), not from equal lengths.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q5/5

Prove the converse of the triangle proportionality theorem

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Use the triangle proportionality theorem	**HSG-SRT.B.4**
BACKWARD 2	SAS similarity	**HSG-SRT.A.3**
BACKWARD 3	Converse of the corresponding angles theorem	**HSG-CO.C.9**
FORWARD 1	Use the converse to decide parallelism in figures	**HSG-SRT.B.4**
FORWARD 2	Prove the trapezoid midsegment theorem	**HSG-SRT.B.5**

SET 43 • HSG-SRT.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with AD/DB = AE/EC. Prove that DE ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q1/5

1745

QUESTION ID

S43-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

DE ∥ BC

GRADING NOTE REQUIRED WORK: Turn the given ratio into AD/AB = AE/AC, prove △ADE ~ △ABC by SAS similarity, then use congruent corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: AD/DB = AE/EC gives AD/AB = AE/AC (add 1 to the reciprocals); ∠A shared; SAS similarity; ∠ADE ≅ ∠ABC; congruent corresponding angles make DE ∥ BC (converse). Invalid: assuming DE ∥ BC to get the similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR, S is on PQ and T is on PR with PS = 4, SQ = 6, PT = 6, TR = 9. Prove that ST ∥ QR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q2/5

1746

QUESTION ID

S43-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

ST ∥ QR

GRADING NOTE REQUIRED WORK: Show PS/PQ = PT/PR, then SAS similarity and corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: PS/PQ = 4/10 = 0.4 and PT/PR = 6/15 = 0.4; ∠P shared; SAS similarity; corresponding angles congruent; converse of corresponding angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with AD/AB = AE/AC. Prove that DE ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q3/5

1747

QUESTION ID

S43-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

DE ∥ BC

GRADING NOTE REQUIRED WORK: SAS similarity using the shared angle, then congruent corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: AD/AB = AE/AC and ∠A shared; SAS similarity gives △ADE ~ △ABC; ∠ADE ≅ ∠ABC; congruent corresponding angles (transversal AB) make DE ∥ BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, M is the midpoint of AB and N is the midpoint of AC. Use the converse of the triangle proportionality theorem to prove that MN ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q4/5

1748

QUESTION ID

S43-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

MN ∥ BC

GRADING NOTE

REQUIRED WORK: Show the two side ratios are equal, then apply the converse.

Any valid proof (paragraph, two-column, or flow). Essential: AM = MB and AN = NC, so AM/MB = 1 = AN/NC; by the converse, MN ∥ BC. Invalid: assuming MN ∥ BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with AD = 3, DB = 5, AE = 4.5, EC = 7.5, and BC = 16. Prove that DE ∥ BC, then find DE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q5/5

1749

QUESTION ID

S43-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

DE ∥ BC; DE = 6

GRADING NOTE REQUIRED WORK: Compare AD/DB and AE/EC, apply the converse, then use △ADE ~ △ABC.

Any valid proof (paragraph, two-column, or flow). Essential: 3/5 = 0.6 = 4.5/7.5, so DE ∥ BC by the converse; △ADE ~ △ABC with ratio AD/AB = 3/8; DE = 16 · 3/8 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 1

Use the triangle proportionality theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC.
AD = 6, DB = 4, AE = 9. Find EC.

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q1/5

1751

QUESTION ID
S43-B1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q1/5

True or false?

In △ABC, D is on AB and E is on AC. If DE ∥ BC, then AD/DB = AE/EC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q2/5

1752

QUESTION ID

S43-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 2, AB = 8, AC = 12. Find AE.

A. 9

B. 4

C. 6

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q3/5

1753

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE 2/8 = AE/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC. State the converse of "If DE ∥ BC, then AD/DB = AE/EC."

Converse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q4/5

1754

QUESTION ID
S43-B1-Q4

TYPE
Short answer

CORRECT ANSWER
If AD/DB = AE/EC, then DE ∥ BC.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q4/5

True or false?

A theorem and its converse are always both true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q5/5

1755

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Some converses are false; this one must be proved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 2

SAS similarity • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two pairs of proportional sides with congruent included angles make triangles similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q1/5

1757

QUESTION ID

S43-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE SAS similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC, so △ADE and △ABC share ∠A. AD/AB = AE/AC = 0.4. Are they similar? Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q2/5

1758

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, SAS similarity

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

After △ADE ~ △ABC, which angle is congruent to ∠ADE?

A. ∠ACB

B. ∠BAC

C. ∠ABC

D. ∠DEA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q3/5

1759

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. ∠ABC

GRADING NOTE D ↔ B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Ratios 3/6 and 4/8 with a shared included angle: similar? Factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q4/5

1760

QUESTION ID

S43-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Yes; factor 2 (or ½)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q4/5

True or false?

SAS similarity needs the angle to be between the two pairs of sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q5/5

1761

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 3

Converse of the corresponding angles theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Congruent corresponding angles make two lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q1/5

1763

QUESTION ID

S43-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC. ∠ADE ≅ ∠ABC, with AB as the transversal. What can you conclude about DE and BC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q2/5

1764

QUESTION ID

S43-B3-Q2

TYPE

Short answer

CORRECT ANSWER

DE ∥ BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q2/5

In △ABC, D is on AB and E is on AC. Which angles, if congruent, prove DE ∥ BC?

A. ∠A and ∠B

B. ∠B and ∠C

C. ∠AED and ∠ACB

D. ∠ADE and ∠AED

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q3/5

1765

QUESTION ID

S43-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. ∠AED and ∠ACB

GRADING NOTE Corresponding angles along AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q3/5

Corresponding angles measure 58° and 62°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q4/5

1766

QUESTION ID

S43-B3-Q4

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q4/5

True or false?

Vertical angles prove lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q5/5

1767

QUESTION ID

S43-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are always congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSG-SRT.B.4

FORWARD 1

Use the converse to decide parallelism in figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In △ABC, D is on AB with AD = 5, DB = 10, and E is on AC with AE = 4. How long must EC be for DE ∥ BC?

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q1/5

1769

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR, S on PQ, T on PR: PS = x, SQ = 6, PT = 4, TR = 8. For which x is ST ∥ QR?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q2/5

1770

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q2/5

True or false?

The midsegment theorem's parallel part is a special case of the converse (ratio 1 : 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q3/5

1771

QUESTION ID

S43-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE AD/DB = AE/EC = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q3/5

Points divide two sides in ratios 2 : 3 and 4 : 6. The segment joining them is

A. perpendicular to the third side

B. half the third side

C. parallel to the third side

D. not parallel to the third side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q4/5

1772

QUESTION ID

S43-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. parallel to the third side

GRADING NOTE 2 : 3 = 4 : 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB with AD : DB = 1 : 2 and DE ∥ BC with E on AC. Find DE : BC.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q5/5

1773

QUESTION ID
S43-F1-Q5

TYPE
Short answer

CORRECT ANSWER
1 : 3

GRADING NOTE Any equivalent form. AD : AB = 1 : 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

FORWARD 2

Prove the trapezoid midsegment theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Trapezoid ABCD has AB ∥ DC. M and N are the midpoints of legs AD and BC. Prove that MN ∥ AB and MN = ½(AB + DC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

1775

QUESTION ID

S43-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

MN ∥ AB and MN = ½(AB + DC)

GRADING NOTE REQUIRED WORK: Draw diagonal AC; let P be the midpoint of AC; use the midsegment theorem (or the converse) in △ACD and △ABC.

Any valid proof (paragraph, two-column, or flow). Essential: in △ADC, MP ∥ DC and MP = ½DC; in △ABC, PN ∥ AB and PN = ½AB; AB ∥ DC so MP and PN are both parallel to AB through P, so M, P, N are collinear; MN = ½(AB + DC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A trapezoid has bases 10 and 16. Find the length of the segment joining the midpoints of the legs.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

1776

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

True or false?

The midsegment of a trapezoid is half the length of the longer base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

1777

QUESTION ID
S43-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is half the sum of the bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

A trapezoid's midsegment is 12 and one base is 9. Find the other base.

A. 3

B. 15

C. 6

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

1778

QUESTION ID

S43-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 15

GRADING NOTE (9 + b)/2 = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Trapezoid ABCD has AB ∥ DC; M and N are the midpoints of legs AD and BC, and P is the midpoint of diagonal AC. Explain why M, P and N lie on one line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

1779

QUESTION ID
S43-F2-Q5

TYPE
Short answer

CORRECT ANSWER
MP and PN are both parallel to the bases through P, and only one line through P is parallel to them.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

Prove the Pythagorean Theorem using similar triangles

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Show triangles similar with AA	**HSG-SRT.A.3**
BACKWARD 2	Cross-multiply proportions	**7.RP.A.2**
BACKWARD 3	Explain a proof of the Pythagorean Theorem informally	**8.G.B.6**
FORWARD 1	Use the geometric mean relationships in a right triangle	**HSG-SRT.B.5**
FORWARD 2	The Law of Cosines as a generalization of the Pythagorean Theor	**HSG-SRT.D.10**

SET 44 • HSG-SRT.B.4

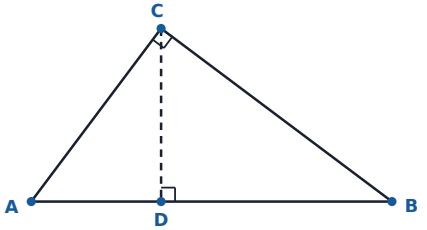
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, legs a = BC, b = AC, and hypotenuse c = AB. The altitude from C meets AB at D. Prove that $a^2 + b^2 = c^2$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q1/5

1782

QUESTION ID

S44-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$a^2 + b^2 = c^2$

GRADING NOTE REQUIRED WORK: Show △ACD ~ △ABC and △CBD ~ △ABC (AA), write two proportions, and add.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △ACB (shared ∠A, right angles), so AD/AC = AC/AB and $b^2 = c \cdot AD$; △CDB ~ △ACB (shared ∠B), so BD/BC = BC/AB and $a^2 = c \cdot BD$; adding, $a^2 + b^2 = c(AD + BD) = c \cdot c = c^2$. Invalid: assuming $a^2 + b^2 = c^2$ anywhere; using the distance formula (it depends on the theorem).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle PQR has the right angle at R, legs p = QR and q = PR, and hypotenuse r = PQ. The altitude from R meets PQ at S. Prove that $p^2 + q^2 = r^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q2/5

1783

QUESTION ID

S44-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$p^2 + q^2 = r^2$

GRADING NOTE REQUIRED WORK: Prove △PSR ~ △PRQ and △RSQ ~ △PRQ (AA), write the two proportions, and add.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠P and right angles give △PSR ~ △PRQ, so PS/PR = PR/PQ and $q^2 = r \cdot PS$; shared ∠Q gives △RSQ ~ △PRQ, so QS/QR = QR/QP and $p^2 = r \cdot QS$; adding, $p^2 + q^2 = r(PS + QS) = r^2$. Invalid: assuming the theorem or using the distance formula.

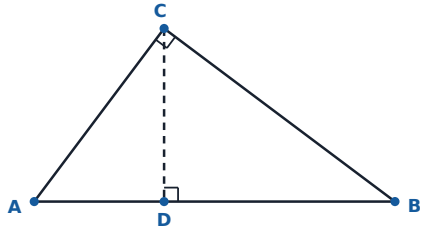
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Isosceles right triangle ABC has the right angle at C and legs AC = BC = a. The altitude from C meets AB at D. Using similar triangles (not the Pythagorean Theorem), prove that $AB^2 = 2a^2$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q3/5

1784

QUESTION ID

S44-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$AB^2 = 2a^2$

GRADING NOTE REQUIRED WORK: Use the similarity proportions $a^2 = AB \cdot AD$ and $a^2 = AB \cdot BD$, then add.

Any valid proof (paragraph, two-column, or flow). Essential: $\triangle ADC \sim \triangle ACB$ gives $a^2 = AB \cdot AD$; $\triangle CDB \sim \triangle ACB$ gives $a^2 = AB \cdot BD$; adding, $2a^2 = AB(AD + BD) = AB^2$.

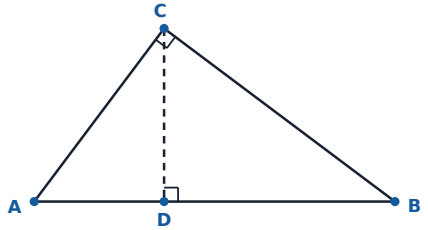
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, AC = 6 and BC = 8. The altitude from C meets AB at D; let x = AD and y = DB. Use similar triangles (not the Pythagorean Theorem) to find x, y and AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q4/5

1785

QUESTION ID

S44-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AB = 10, x = 3.6, y = 6.4

GRADING NOTE REQUIRED WORK: Write $6^2 = AB \cdot x$ and $8^2 = AB \cdot y$ from the similar triangles, and use $x + y = AB$.

Any valid proof (paragraph, two-column, or flow). Essential: $\triangle ADC \sim \triangle ACB$ gives $36 = AB \cdot x$; $\triangle CDB \sim \triangle ACB$ gives $64 = AB \cdot y$; adding, $100 = AB(x + y) = AB^2$, so AB = 10, x = 3.6, y = 6.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 5 and 12. The altitude to the hypotenuse divides it into parts x (next to the leg 5) and y. Use similar triangles (not the Pythagorean Theorem) to find the hypotenuse c, x and y.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q5/5

1786

QUESTION ID

S44-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

c = 13, x = 25/13, y = 144/13

GRADING NOTE

REQUIRED WORK: Write $5^2 = c \cdot x$ and $12^2 = c \cdot y$, add, and use $x + y = c$.

Any valid proof (paragraph, two-column, or flow). Essential: $25 = cx$ and $144 = cy$; adding, $169 = c(x + y) = c^2$, so $c = 13$; $x = 25/13 \approx 1.92$ and $y = 144/13 \approx 11.08$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 1

Show triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two right triangles that share an acute angle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q1/5

1788

QUESTION ID
S44-B1-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, and the altitude from C meets hypotenuse AB at D. Name two pairs of congruent angles in △ADC and △ACB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q2/5

1789

QUESTION ID
S44-B1-Q2

TYPE
Short answer

CORRECT ANSWER
∠A ≅ ∠A; ∠ADC ≅ ∠ACB (right angles)

GRADING NOTE Any two valid pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q2/5

Right triangle ABC has the right angle at C, and the altitude from C meets hypotenuse AB at D. Then △ADC ~ △ACB. Which ratio equals AD/AC?

A. DC/AB

B. AD/AB

C. AC/AB

D. CB/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q3/5

1790

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. AC/AB

GRADING NOTE AD ↔ AC, AC ↔ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, and the altitude from C meets hypotenuse AB at D. △ADC ~ △ACB, AD = 4 and AB = 9. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q4/5

1791

QUESTION ID
S44-B1-Q4

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form. $AC^2 = AD \cdot AB = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q4/5

True or false?

Similar triangles must be the same size.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q5/5

1792

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only the shape matches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 2

Cross-multiply proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From AD/AC = AC/AB, write an equation without fractions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

1794

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$AC^2 = AD \cdot AB$

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $4/x = x/9$ for $x > 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

1795

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If $p/q = q/r$, then

A. $q^2 = pr$

B. $p = r$

C. $r^2 = pq$

D. $p^2 = qr$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

1796

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $q^2 = pr$

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From $a/b = b/c$ we get $b^2 = ac$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q4/5

1797

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From x/5 = 5/x we get x = 5 only.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q5/5

1798

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = 25$, so $x = \pm 5$ ($x = 5$ for a length).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 3

Explain a proof of the Pythagorean Theorem informally • Grade 8 • Geometry • 8.G.B.6

True or false?

The Pythagorean Theorem says $a^2 + b^2 = c^2$ for every right triangle with legs a, b and hypotenuse c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

1800

QUESTION ID

S44-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name a way to show the theorem by rearranging area pieces.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

1801

QUESTION ID

S44-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Arrange four copies of the triangle inside a square of side a + b (the leftover is c^2, or $a^2 + b^2$).

GRADING NOTE Any valid area argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

Which triangle satisfies the theorem?

A. 5, 12, 13

B. 6, 7, 9

C. 5, 6, 7

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

1802

QUESTION ID

S44-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5, 12, 13

GRADING NOTE 25 + 144 = 169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 9 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

1803

QUESTION ID

S44-B3-Q4

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

True or false?

Checking the theorem with three triangles proves it for all right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

1804

QUESTION ID

S44-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Examples are not proofs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 1

Use the geometric mean relationships in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, and altitude CD meets hypotenuse AB at D. AD = 4 and DB = 9. Find CD.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

1806

QUESTION ID
S44-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form. $CD^2 = AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with the right angle at C, altitude CD meets hypotenuse AB at D. Prove that $CD^2 = AD \cdot DB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5

1807

QUESTION ID

S44-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

$CD^2 = AD \cdot DB$

GRADING NOTE REQUIRED WORK: Show △ACD ~ △CBD and write a proportion.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △CDB (AA: right angles at D, and ∠A ≅ ∠BCD since both are complementary to ∠B); AD/CD = CD/DB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5

True or false?

The altitude to the hypotenuse is the geometric mean of the two parts of the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

1808

QUESTION ID

S44-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $CD^2 = AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C, and the altitude from C meets hypotenuse AB at D. AD = 2 and AB = 8. Find AC.

A. 4

B. $2\sqrt{2}$

C. 6

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

1809

QUESTION ID

S44-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $AC^2 = AD \cdot AB = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle's altitude to the hypotenuse is 12 and one part of the hypotenuse is 9. Find the other part.

Part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

1810

QUESTION ID

S44-F1-Q5

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form. 144 = 9x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 2

The Law of Cosines as a generalization of the Pythagorean Theorem • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

For a triangle with sides a, b, c and angle C opposite c, the Law of Cosines says $c^2 = a^2 + b^2 - 2ab \cos C$. What does it become when $C = 90°$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5

1812

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$c^2 = a^2 + b^2$ (the Pythagorean Theorem), since $\cos 90° = 0$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides 5 and 8 with a 60° angle between them. Find the third side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

1813

QUESTION ID
S44-F2-Q2

TYPE
Short answer

CORRECT ANSWER
7

GRADING NOTE Any equivalent form. 25 + 64 − 40 = 49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

True or false?

If $a^2 + b^2 > c^2$, the angle opposite c is obtuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5

1814

QUESTION ID

S44-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is acute; the angle is obtuse when $a^2 + b^2 < c^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5

A triangle has sides 4, 5, 7. The largest angle is

A. right

B. impossible to tell

C. obtuse

D. acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5

1815

QUESTION ID

S44-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. obtuse

GRADING NOTE $16 + 25 = 41 < 49$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5

Use the converse idea: is a triangle with sides 7, 24, 25 right, acute, or obtuse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5

1816

QUESTION ID

S44-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Right

(49 + 576 = 625)

GRADING NOTE Form required: the classification with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5

Use similarity criteria to solve problems in geometric figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Similarity criteria: AA, SAS, SSS	**HSG-SRT.A.3**
BACKWARD 2	Solve proportions	**7.RP.A.2**
BACKWARD 3	Identify corresponding sides in overlapping triangles	**HSG-SRT.A.2**
FORWARD 1	Solve multistep problems with two or more pairs of similar triangl...	**HSG-SRT.B.5**
FORWARD 2	Solve similarity problems with algebra (unknowns on both sides)	**HSG-SRT.B.5**

SET 45 • HSG-SRT.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 6, DB = 9, DE = 8. Find BC. Name the criterion you used for similar triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

1819

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

BC = 20; △ADE ~ △ABC by AA.

GRADING NOTE Form required: the length and the criterion. 6/15 = 8/BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A lamppost lights a student who is 1.6 m tall and stands 3 m from the post. The student's shadow is 2 m long, measured away from the post. How tall is the lamppost? Name the similarity criterion you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

1820

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 m; AA (the shared angle at the tip of the shadow, and two right angles).

GRADING NOTE Form required: the height and the criterion. The tip of the shadow is 5 m from the post: 1.6/2 = h/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

Two triangles: △PQR with PQ = 8, QR = 12, ∠Q = 50°; △STU with ST = 6, TU = 9, ∠T = 50°. Are they similar? If so, find the ratio of their perimeters.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

1821

QUESTION ID

S45-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes, SAS similarity (8/6 = 12/9 = 4/3); perimeter ratio 4 : 3.

GRADING NOTE Form required: the criterion and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To find the width w of a river, a surveyor marks points so that segments AD and BE cross at C, with ∠B and ∠E right angles. AB = 12 m, BC = 18 m and CE = 60 m; w = DE is across the river. Find w and name the similarity criterion you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

1822

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

w = 40 m; △ABC ~ △DEC by AA (right angles at B and E, vertical angles at C).

GRADING NOTE Form required: the width and the criterion. 12/18 = w/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

△KLM has KL = 6, LM = 8 and KM = 12, and m∠L ≈ 117°. △NOP has NO = 9, OP = 12 and NP = 18. Are the triangles similar? Name the criterion, give the scale factor, and find m∠O.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

1823

QUESTION ID

S45-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes, SSS similarity with scale factor 1.5 (9/6 = 12/8 = 18/12); ∠O corresponds to ∠L, so m∠O ≈ 117°.

GRADING NOTE Form required: the criterion, the factor, and the angle with the correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 1

Similarity criteria: AA, SAS, SSS • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which criterion uses two pairs of congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5

1825

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

AA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Sides 2, 3, 4 and 4, 6, 8. Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

1826

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

SSS similarity

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

Which information is enough for similarity?

A. Two pairs of congruent angles

B. One pair of congruent angles

C. One pair of proportional sides

D. Two sides of each triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

1827

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Two pairs of congruent angles

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

True or false?

SAS similarity requires the congruent angle to be between the proportional sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

1828

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

1829

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They need another congruent angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 6/15 = 8/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

1831

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 1.6/2 = h/5.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

1832

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 12/40 = 18/x.

A. 26.7

B. 54

C. 46

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

1833

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60

GRADING NOTE x = 18 · 40/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

3/4 = 9/12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

1834

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5/8 = 10/14

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

1835

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10/16 would be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 3

Identify corresponding sides in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Which side of △ABC corresponds to DE?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

1837

QUESTION ID

S45-B3-Q1

TYPE

Short answer

CORRECT ANSWER

BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

True or false?

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Side AD corresponds to side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

1838

QUESTION ID
S45-B3-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. In △ADE ~ △ABC, the side of the large triangle that corresponds to AD is

A. DB

B. AC

C. BC

D. AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

1839

QUESTION ID

S45-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. AB

GRADING NOTE Whole side, not the piece DB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

D is on segment AB with AD = 6 and DB = 9. What is AB?

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5

1840

QUESTION ID

S45-B3-Q4

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B3-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. DB corresponds to DE.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q5/5

1841

QUESTION ID
S45-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE DB is not a side of △ADE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSG-SRT.B.5

FORWARD 1

Solve multistep problems with two or more pairs of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Trapezoid ABCD has AB ∥ DC, AB = 15, DC = 10. The diagonals meet at E, and AC = 20. Find AE and EC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5

1843

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

AE = 12, EC = 8

GRADING NOTE Form required: both. AE/EC = 15/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pinhole camera: an object 2 m tall is 8 m from the pinhole; the image is 5 cm from the pinhole on the screen. How tall is the image?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

1844

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.25 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

True or false?

In a figure with two pairs of similar triangles, one ratio found in the first pair can be reused in the second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

1845

QUESTION ID

S45-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Shared sides link them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D and F are on AB and E and G are on AC, with DE ∥ BC and FG ∥ BC. AD = 2, DB = 6, AF = 5 and BC = 16. Find FG.

A. 10

B. 12

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

1846

QUESTION ID

S45-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE FG/BC = AF/AB = 5/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Using a mirror on the ground, Ana (eyes 1.5 m high) stands 2 m from the mirror and sees the top of a tree 12 m beyond the mirror. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

1847

QUESTION ID
S45-F1-Q5

TYPE
Short answer

CORRECT ANSWER
9 m

GRADING NOTE Any equivalent form. 1.5/2 = h/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

FORWARD 2

Solve similarity problems with algebra (unknowns on both sides) • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △ABC, D is on AB and E is on AC with DE ∥ BC.
AD = x, DB = 6, AE = 4, EC = x + 2. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

1849

QUESTION ID
S45-F2-Q1

TYPE
Short answer

CORRECT ANSWER
4

GRADING NOTE Any equivalent form. x(x + 2) = 24, x > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

Similar triangles have corresponding sides x + 1 and 2x − 3 with scale factor 1.5 (larger to smaller is 2x − 3 to x + 1). Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5

1850

QUESTION ID
S45-F2-Q2

TYPE
Short answer

CORRECT ANSWER
9

GRADING NOTE Any equivalent form. 2x − 3 = 1.5(x + 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5

True or false?

A proportion with the unknown in two places can lead to a quadratic equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

1851

QUESTION ID

S45-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cross-multiplying gives products.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

In right triangle ABC with the right angle at C, the altitude CD meets hypotenuse AB at D. AD = x, DB = x + 5, and CD = 6. Find x.

A. 4

B. 3

C. 6

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

1852

QUESTION ID

S45-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE x(x + 5) = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two similar rectangles: the first is 4 by x, the second is x by 9. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5

1853

QUESTION ID

S45-F2-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 4/x = x/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5

Use congruence criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Choose and apply SSS, SAS, ASA, AAS	**HSG-CO.B.8**
BACKWARD 2	Find hidden parts: shared sides, vertical angles, right angles	**HSG-CO.C.9**
BACKWARD 3	Use corresponding parts to finish a proof	**HSG-CO.B.7**
FORWARD 1	Prove relationships that need two congruences in a row	**HSG-SRT.B.5**
FORWARD 2	Decide which given data are sufficient for a proof, and supply a m…	**HSG-SRT.B.5**

SET 46 • HSG-SRT.B.5

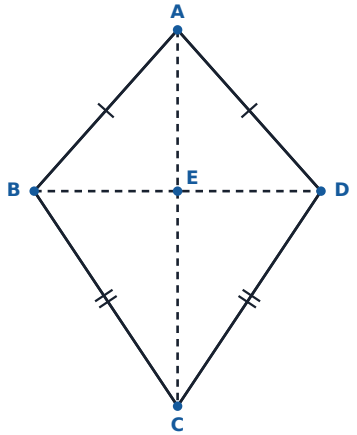
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Kite ABCD has AB = AD and CB = CD. Its diagonals meet at E. Prove that AC ⊥ BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

1856

QUESTION ID

S46-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AC ⊥ BD

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △ADC (SSS) to get ∠BAE ≅ ∠DAE; then △ABE ≅ △ADE (SAS); then the angles at E form a congruent linear pair.

Any valid proof (paragraph, two-column, or flow). Essential: SSS (AB = AD, CB = CD, AC shared); ∠BAC ≅ ∠DAC; SAS on △ABE and △ADE (AE shared); ∠AEB ≅ ∠AED, a linear pair, so 90°. Alternative: A and C are each equidistant from B and D, so AC is the perpendicular bisector of BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

Segments AB and CD bisect each other at M. Prove that AC ∥ DB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

1857

QUESTION ID

S46-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

AC ∥ DB

GRADING NOTE REQUIRED WORK: Prove △AMC ≅ △BMD (SAS with vertical angles), then use congruent alternate interior angles.

Any valid proof (paragraph, two-column, or flow). Essential: AM = BM, CM = DM (bisect), ∠AMC ≅ ∠BMD (vertical); SAS; ∠CAM ≅ ∠DBM; these are alternate interior angles for AC and DB with transversal AB; converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Point P is on the bisector of ∠ABC. PD ⊥ BA at D and PE ⊥ BC at E. Prove that PD = PE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

1858

QUESTION ID

S46-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

PD = PE

GRADING NOTE REQUIRED WORK: Prove △PBD ≅ △PBE by AAS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PBD ≅ ∠PBE (bisector), ∠PDB ≅ ∠PEB (right angles), PB shared; AAS; PD = PE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, AB = AC. M is the midpoint of BC. Prove that AM bisects ∠BAC and AM ⊥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

1859

QUESTION ID

S46-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

∠BAM ≅ ∠CAM and AM ⊥ BC

GRADING NOTE REQUIRED WORK: Prove △ABM ≅ △ACM (SSS), then use corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: SSS; ∠BAM ≅ ∠CAM (bisects); ∠AMB ≅ ∠AMC form a linear pair, so each is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, AB ∥ DC and AB = DC. The diagonals AC and BD meet at E. Prove that the diagonals bisect each other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

1860

QUESTION ID

S46-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

AE = CE and BE = DE

GRADING NOTE REQUIRED WORK: Prove △ABE ≅ △CDE by ASA using alternate interior angles, then use corresponding parts.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAE ≅ ∠DCE and ∠ABE ≅ ∠CDE (alternate interior angles, AB ∥ DC); AB = CD; ASA; AE = CE and BE = DE (corresponding parts).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 1

Choose and apply SSS, SAS, ASA, AAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles share a side and have two more pairs of congruent sides. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

1862

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles and a non-included side match. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

1863

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

AAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

Which pair of facts plus a shared side gives SAS?

A. A pair of congruent sides and the congruent included angles

B. Two pairs of congruent sides not including the angle

C. Nothing more is needed

D. Two pairs of congruent angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

1864

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A pair of congruent sides and the congruent included angles

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

True or false?

A shared side is a valid pair for any criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

1865

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

True or false?

AAA proves congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

1866

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 2

Find hidden parts: shared sides, vertical angles, right angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Segments AB and CD cross at M. Which angles are vertical to ∠AMC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

1868

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

∠BMD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

True or false?

Perpendicular lines give congruent right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

1869

QUESTION ID

S46-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

Which is a hidden fact in figures where two triangles share a side?

A. All angles are equal

B. The shared side is congruent to itself

C. The triangles are right

D. Nothing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

1870

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The shared side is congruent to itself

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A bisector divides ∠ABC into which two congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

1871

QUESTION ID

S46-B2-Q4

TYPE

Short answer

CORRECT ANSWER

∠ABP and ∠PBC (for bisector ray BP)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

True or false?

Vertical angles are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

1872

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 3

Use corresponding parts to finish a proof • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

ANSWER KEY Reduced copy of the question page, with its answer

△ABC ≅ △ADC. Which angle corresponds to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

1874

QUESTION ID

S46-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠DAC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

True or false?

Once triangles are congruent, all corresponding parts are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

1875

QUESTION ID

S46-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△PBD ≅ △PBE. Which conclusion follows?

A. PD = PE

B. PB = PD

C. BD = PE

D. ∠D ≅ ∠B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

1876

QUESTION ID

S46-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. PD = PE

GRADING NOTE Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two congruent angles form a linear pair. Each measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

1877

QUESTION ID

S46-B3-Q4

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

True or false?

Corresponding parts can be used before the triangles are proved congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

1878

QUESTION ID

S46-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only after.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 1

Prove relationships that need two congruences in a row • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, AB = CD and AD = CB. Diagonals AC and BD meet at E. Prove that E is the midpoint of AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

1880

QUESTION ID

S46-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AE = CE

GRADING NOTE

REQUIRED WORK: Prove △ABC ≅ △CDA (SSS) to get parallel sides, then △ABE ≅ △CDE (ASA).

Any valid proof (paragraph, two-column, or flow). Essential: SSS gives alternate interior angles congruent (AB ∥ DC); then ASA (or AAS) for △ABE and △CDE gives AE = CE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, AB = AC. D is on AB and E is on AC with BD = CE. Prove that BE = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

1881

QUESTION ID

S46-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

BE = CD

GRADING NOTE REQUIRED WORK: Show AD = AE, then prove △ABE ≅ △ACD by SAS (shared ∠A).

Any valid proof (paragraph, two-column, or flow). Essential: AD = AB − BD = AC − CE = AE; ∠A shared; AB = AC; SAS; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

True or false?

Some proofs need one congruence to find a part used in a second congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

1882

QUESTION ID

S46-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Chained congruences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

In a two-step proof, the first congruence gives $\angle 1 \cong \angle 2$. This is used to

A. measure

B. end the proof

C. supply a pair of parts for the second congruence

D. restart

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

1883

QUESTION ID

S46-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. supply a pair of parts for the second congruence

GRADING NOTE Chaining.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why you cannot use corresponding parts of the second pair of triangles before proving them congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

1884

QUESTION ID
S46-F1-Q5

TYPE
Short answer

CORRECT ANSWER
That would assume what must be proved (circular reasoning).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 2

Decide which given data are sufficient for a proof, and supply a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR and △PSR (sharing PR), PQ = PS. What one more fact would prove ∠QPR ≅ ∠SPR? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

1886

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

QR = SR (then SSS gives the congruence and the angles correspond).

GRADING NOTE Any sufficient condition with explanation (∠PRQ ≅ ∠PRS would give SSA, which is not enough).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

A student claims a kite's diagonals bisect each other. Prove or disprove.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

1887

GRADING NOTE Any valid counterexample or explanation.

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

False in general: only the diagonal BD is bisected (by AC); AC is bisected only if the kite is a rhombus. Example: A(0, 3), B(−1, 0), C(0, −1), D(1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

True or false?

Given AB = DE, BC = EF, and ∠C ≅ ∠F, the triangles must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

1888

QUESTION ID

S46-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE SSA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

Given ∠A ≅ ∠D and AB = DE in △ABC and △DEF, which extra fact gives ASA?

A. ∠C ≅ ∠F

B. BC = EF

C. AC = DF

D. ∠B ≅ ∠E

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

1889

QUESTION ID

S46-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ∠B ≅ ∠E

GRADING NOTE The angle at the other end of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Given AB = DE and AC = DF, name an angle condition that guarantees congruence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

1890

QUESTION ID
S46-F2-Q5

TYPE
Short answer

CORRECT ANSWER
∠A ≅ ∠D (the included angle)

GRADING NOTE Form required: the included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

Use similarity criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Prove triangles similar with AA	**HSG-SRT.A.3**
BACKWARD 2	Turn a proportion into a product	**7.RP.A.2**
BACKWARD 3	Match corresponding sides of similar triangles	**HSG-SRT.A.2**
FORWARD 1	Use proved product relationships to compute	**HSG-SRT.B.5**
FORWARD 2	Prove relationships in circles with similar triangles	**HSG-C.A.2**

SET 47 • HSG-SRT.B.5

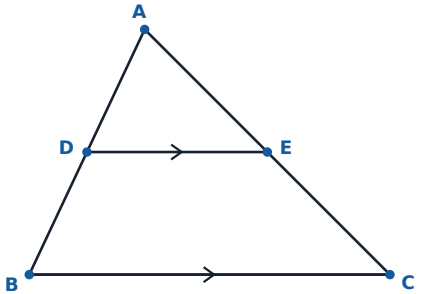
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC with D on AB and E on AC. Prove that AD · AC = AE · AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

1893

QUESTION ID

S47-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AD · AC = AE · AB

GRADING NOTE REQUIRED WORK: Prove △ADE ~ △ABC (AA), write the proportion, and cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: AA (shared ∠A; corresponding angles); AD/AB = AE/AC; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Chords AB and CD of a circle meet at P, and ∠BAD ≅ ∠BCD (given). Prove that PA · PB = PC · PD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

1894

QUESTION ID

S47-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

PA · PB = PC · PD

GRADING NOTE REQUIRED WORK: Prove △PAD ~ △PCB (AA: the given angles and vertical angles), then cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PAD ≅ ∠PCB (given), ∠APD ≅ ∠CPB (vertical); AA; PA/PC = PD/PB; cross-multiply.

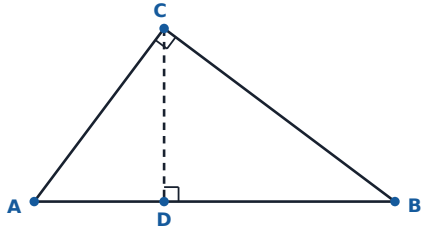
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), CD is the altitude to AB. Prove that AC · BC = AB · CD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

1895

QUESTION ID

S47-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

AC · BC = AB · CD

GRADING NOTE REQUIRED WORK: Prove △ACB ~ △ADC (or △CDB) and use the ratio of corresponding sides.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △ACB (AA); DC/CB = AC/AB; cross-multiply. Alternative: both products equal twice the area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In trapezoid ABCD with AB ∥ DC, the diagonals AC and BD meet at E. Prove that AE/EC = AB/DC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

1896

QUESTION ID

S47-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AE/EC = AB/DC

GRADING NOTE REQUIRED WORK: Prove △ABE ~ △CDE by AA (alternate interior angles or vertical angles).

Any valid proof (paragraph, two-column, or flow). Essential: ∠EAB ≅ ∠ECD and ∠EBA ≅ ∠EDC (alternate interior, AB ∥ DC); AA; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In acute triangle ABC, the altitudes from A and B meet BC at D and AC at E. Prove that CD · CB = CE · CA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

1897

QUESTION ID

S47-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

CD · CB = CE · CA

GRADING NOTE REQUIRED WORK: Prove △CDA ~ △CEB (AA), write the proportion, and cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: ∠C is shared; ∠CDA and ∠CEB are right angles; AA; CD/CE = CA/CB; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 1

Prove triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Vertical angles can be one pair of congruent angles in an AA proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

1899

QUESTION ID
S47-B1-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles share ∠A and have another pair of congruent angles. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5

1900

QUESTION ID
S47-B1-Q2

TYPE
Short answer

CORRECT ANSWER
AA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5

Parallel lines in a figure usually supply

A. equal side lengths

B. a scale factor

C. right angles

D. congruent corresponding or alternate interior angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q3/5

1901

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. congruent corresponding or alternate interior angles

GRADING NOTE Angle pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B1-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

△PAD ~ △PCB. Which side of △PCB corresponds to PA?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5

1902

QUESTION ID

S47-B1-Q4

TYPE

Short answer

CORRECT ANSWER

PC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Similarity requires all three pairs of sides to be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

1903

QUESTION ID
S47-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Proportional, not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 2

Turn a proportion into a product • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From a/b = c/d, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

1905

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

ad = bc

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From PA/PC = PD/PB, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

1906

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

PA · PB = PC · PD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If x/3 = 4/6, then

A. 6x = 12

B. x = 8

C. 3x = 24

D. 4x = 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

1907

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6x = 12

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

a/b = c/d implies ab = cd.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

1908

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It implies ad = bc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

2/5 = 6/15 implies 2 · 15 = 5 · 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q5/5

1909

QUESTION ID

S47-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 30 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 3

Match corresponding sides of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC, and △ADE ~ △ABC. Write the proportion with AD, AB, AE, AC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

1911

QUESTION ID

S47-B3-Q1

TYPE

Short answer

CORRECT ANSWER

AD/AB = AE/AC

GRADING NOTE Any equivalent proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

True or false?

In △ABE ~ △CDE, AB corresponds to CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

1912

QUESTION ID

S47-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A ↔ C, B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△ACB ~ △ADC. Which side corresponds to CB?

A. DC

B. AD

C. AB

D. AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5

1913

QUESTION ID

S47-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. DC

GRADING NOTE A ↔ A, C ↔ D, B ↔ C, so CB ↔ DC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B3-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

△XYZ ~ △UVW. Write all three equal ratios.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

1914

QUESTION ID

S47-B3-Q4

TYPE

Short answer

CORRECT ANSWER

XY/UV = YZ/VW = XZ/UW

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

True or false?

Any two sides of similar triangles are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

1915

QUESTION ID

S47-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 1

Use proved product relationships to compute • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Chords AB and CD meet at P with PA = 4, PB = 9, PC = 6. Find PD.

PD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

1917

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 4 · 9 = 6 · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), AC = 6, BC = 8, AB = 10. Find the altitude CD to AB.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

1918

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.8

GRADING NOTE Any equivalent form. 6 · 8 = 10 · CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

True or false?

If two chords meet inside a circle, the products of their segments are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

1919

QUESTION ID
S47-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Proved by similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Trapezoid diagonals: AB = 18, DC = 12, AE = 9. Find EC.

A. 8

B. 4

C. 13.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

1920

QUESTION ID

S47-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE AE/EC = 18/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC.
AD = 3, AB = 9, AC = 12. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

1921

QUESTION ID
S47-F1-Q5

TYPE
Short answer

CORRECT ANSWER
4

GRADING NOTE Any equivalent form. AD · AC = AE · AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 2

Prove relationships in circles with similar triangles • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From point P outside a circle, a tangent touches the circle at T and a secant meets it at A and B (A between P and B). Given ∠PTA ≅ ∠PBT, prove that $PT^2 = PA \cdot PB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

1923

QUESTION ID

S47-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$PT^2 = PA \cdot PB$

GRADING NOTE REQUIRED WORK: Prove △PTA ~ △PBT (AA), then cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠P and the given angles; AA; PT/PB = PA/PT; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

A tangent segment from P is 8 and the secant from P has outer part 4. Find the whole secant length.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

1924

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form. 64 = 4 · PB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

True or false?

Two secants from an outside point P satisfy PA + PB = PC + PD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

1925

QUESTION ID

S47-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The products are equal: PA · PB = PC · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

From P, secants meet a circle at A, B and at C, D with PA = 3, PB = 12, PC = 4. Find PD.

A. 7

B. 1

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

1926

QUESTION ID

S47-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9

GRADING NOTE 36 = 4 · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

Explain why the angle facts used in these circle proofs depend on inscribed angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

1927

QUESTION ID
S47-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Angles that intercept the same arc are congruent (inscribed angle theorem), which supplies the AA pairs.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

Understand trigonometric ratios as properties of an angle, by similarity

MAIN	Geometry main question	**HSG-SRT.C.6**
BACKWARD 1	Similar triangles have equal side ratios within each triangle	**HSG-SRT.A.2**
BACKWARD 2	Name the opposite side, adjacent side, and hypotenuse for an ang	**HSG-SRT.A.3**
BACKWARD 3	Write and simplify ratios	**6.RP.A.1**
FORWARD 1	Find trig ratios from one ratio, using a similar right triangle	**HSG-SRT.C.6**
FORWARD 2	The unit circle extends the ratios beyond acute angles	**HSF-TF.A.2**

SET 48 • HSG-SRT.C.6

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Right triangles ABC and DEF have right angles at C and F and ∠A ≅ ∠D. Explain why BC/AB = EF/DE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

1930

QUESTION ID

S48-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

BC/AB = EF/DE

GRADING NOTE

REQUIRED WORK: Use AA similarity, then the proportion of corresponding sides, rearranged.

Any valid proof (paragraph, two-column, or flow). Essential: right angles and ∠A ≅ ∠D give AA similarity; BC/EF = AB/DE; rearranging gives BC/AB = EF/DE. So the ratio depends only on the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two right triangles each have a 30° angle. In one, the side opposite 30° is 4 and the hypotenuse is 8. In the other, the hypotenuse is 15. Find the side opposite 30°, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

1931

QUESTION ID
S48-M-Q2

TYPE
Short answer

CORRECT ANSWER
7.5; the triangles are similar (AA), so opposite/hypotenuse = 4/8 = ½ in both.

GRADING NOTE Form required: the length and the similarity reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle PQR, the right angle is at R, PR = 5, QR = 12, PQ = 13. Write sin P, cos P, and tan P.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

1932

QUESTION ID

S48-M-Q3

TYPE

Short answer

CORRECT ANSWER

sin P = 12/13, cos P = 5/13, tan P = 12/5

GRADING NOTE Form required: all three ratios (any equivalent fractions).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, AC = 4 and BC = 3. Right triangle DEF has the right angle at F, ∠D ≅ ∠A and DE = 20. Find EF and DF, and explain why tan A = tan D.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

1933

QUESTION ID

S48-M-Q4

TYPE

Short answer

CORRECT ANSWER

AB = 5 and the triangles are similar (AA) with factor 4, so EF = 12 and DF = 16; tan A = 3/4 and tan D = 12/16 = 3/4, equal because corresponding sides are proportional.

GRADING NOTE Form required: both lengths and the similarity explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A student says cos 50° depends on how big the right triangle is. Use two right triangles with a 50° angle, one with hypotenuse 1 and one with hypotenuse 10, to explain why the student is wrong, and give cos 50° to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

1934

QUESTION ID

S48-M-Q5

TYPE

Short answer

CORRECT ANSWER

The triangles are similar (AA), so the side adjacent to 50° is 10 times as long when the hypotenuse is 10 times as long; adjacent/hypotenuse is the same, cos 50° ≈ 0.643.

GRADING NOTE Form required: the similarity explanation and the value to the nearest thousandth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 1

Similar triangles have equal side ratios within each triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Triangles with sides 3, 4, 5 and 6, 8, 10 are similar.
Compare 3/5 and 6/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

1936

QUESTION ID
S48-B1-Q1

TYPE
Short answer

CORRECT ANSWER
Both equal 0.6.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

True or false?

If △ABC ~ △DEF, then AB/AC = DE/DF.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

1937

QUESTION ID

S48-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Ratios within each triangle match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△ABC ~ △DEF with BC/AB = 0.8. Find EF/DE.

A. Cannot tell

B. 0.8

C. 1.25

D. 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

1938

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.8

GRADING NOTE Same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has legs 2 and 3. A similar one has shorter leg 8. Find the longer leg.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

1939

QUESTION ID
S48-B1-Q4

TYPE
Short answer

CORRECT ANSWER
12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Similar triangles have equal side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

1940

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Proportional, not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 2

Name the opposite side, adjacent side, and hypotenuse for an angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C, which side is opposite ∠A?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

1942

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C, which side is adjacent to ∠A (not the hypotenuse)?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

1943

QUESTION ID
S48-B2-Q2

TYPE
Short answer

CORRECT ANSWER
AC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

In a right triangle, the hypotenuse is always

A. opposite the right angle

B. next to the right angle

C. the shortest side

D. opposite the smallest angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

1944

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. opposite the right angle

GRADING NOTE Longest side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

True or false?

In right triangle ABC with the right angle at C, the side opposite ∠A is adjacent to ∠B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

1945

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE BC is next to ∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

True or false?

The opposite side of an angle depends on the triangle's size.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

1946

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It depends on position.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 3

Write and simplify ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 12 : 13 as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

1948

QUESTION ID

S48-B3-Q1

TYPE

Short answer

CORRECT ANSWER

12/13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 8/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

1949

QUESTION ID

S48-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4/5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equals 0.75?

A. 7/5

B. 3/7

C. 3/4

D. 4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

1950

QUESTION ID

S48-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/4

GRADING NOTE 3 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5/13 ≈ 0.385

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

1951

QUESTION ID

S48-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A ratio changes when both quantities are tripled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

1952

QUESTION ID

S48-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 1

Find trig ratios from one ratio, using a similar right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

For an acute angle θ, sin θ = 3/5. Find cos θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

1954

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

cos θ = 4/5, tan θ = 3/4

GRADING NOTE Form required: both. A 3-4-5 triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

tan θ = 5/12 for an acute angle θ. Find sin θ.

sin θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

1955

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5/13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

True or false?

For any acute angle, $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

1956

QUESTION ID

S48-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem divided by the hypotenuse squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = 0.6. Which is sin θ (θ acute)?

A. 0.8

B. 1.6

C. 0.4

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

1957

QUESTION ID

S48-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.8

GRADING NOTE √(1 − 0.36).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\sin\theta < 1$ for every acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

1958

QUESTION ID

S48-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The opposite leg is shorter than the hypotenuse.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 2

The unit circle extends the ratios beyond acute angles • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

On a unit circle, the point for angle θ is (cos θ, sin θ).
Find the point for 90°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5

1960

QUESTION ID
S48-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(0, 1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the coordinates of the point on the unit circle at 30°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5

1961

QUESTION ID
S48-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(√3/2, ½)

GRADING NOTE Form required: exact values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5

True or false?

For acute angles, the unit-circle definition of sine gives a different value from the right-triangle definition.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

1962

QUESTION ID
S48-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They agree: the hypotenuse is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

sin 150° on the unit circle equals

A. −½

B. −√3/2

C. √3/2

D. ½

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5

1963

QUESTION ID

S48-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ½

GRADING NOTE Point (−√3/2, ½).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5

Explain why the unit circle lets us define sine for 120°, when no right triangle has a 120° acute angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

1964

GRADING NOTE Any equivalent answer.

QUESTION ID

S48-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The point on the unit circle at 120° has a y-coordinate, which defines sin 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

Explain and use the relationship between the sine and cosine of complementary angles

MAIN	Geometry main question	**HSG-SRT.C.7**
BACKWARD 1	Complementary angles add to 90°	**7.G.B.5**
BACKWARD 2	Opposite and adjacent sides depend on the chosen angle	**HSG-SRT.C.6**
BACKWARD 3	Angle sum in a right triangle	**8.G.A.5**
FORWARD 1	Relationships between tangents of complementary angles	**HSG-SRT.C.7**
FORWARD 2	The Pythagorean identity	**HSF-TF.C.8**

SET 49 • HSG-SRT.C.7

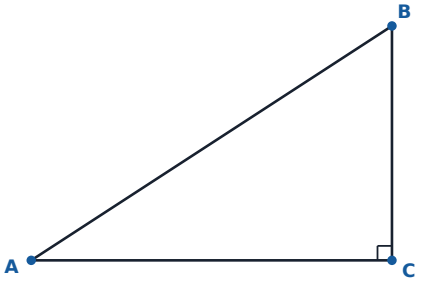
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C. Explain why sin A = cos B.

Work space



Conclusion: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

1967

QUESTION ID

S49-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

sin A = BC/AB = cos B, and A and B are complementary.

GRADING NOTE REQUIRED WORK: Name the side opposite ∠A and the side adjacent to ∠B, and use the angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = BC/AB (opposite over hypotenuse); cos B = BC/AB (BC is adjacent to ∠B); so they are equal; ∠A + ∠B = 90°, so sin A = cos(90° − A).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

sin 25° = cos x°, where 0 < x < 90. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

1968

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

65

GRADING NOTE Any equivalent form. Complementary angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

θ is acute and cos θ = 0.6. Find sin(90° − θ).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

1969

QUESTION ID
S49-M-Q3

TYPE
Short answer

CORRECT ANSWER
0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Both angles are acute and sin(2x + 10)° = cos(3x)°.
Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

1970

QUESTION ID
S49-M-Q4

TYPE
Short answer

CORRECT ANSWER
16

GRADING NOTE Any equivalent form. 2x + 10 + 3x = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A ladder leans against a vertical wall on level ground and makes an acute angle θ with the ground, with sin θ = 0.8. Find cos(90° − θ), say which angle in the picture measures 90° − θ, and explain why the two values are equal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

1971

QUESTION ID

S49-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.8. The angle between the ladder and the wall is 90° − θ; for it, the height on the wall is the adjacent side, and for θ it is the opposite side, so both ratios are height/ladder.

GRADING NOTE Form required: the value, the angle, and the explanation with the shared side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 1

Complementary angles add to 90° • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find the complement of 37°.

Complement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

1973

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

53°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

True or false?

The two acute angles of a right triangle are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

1974

QUESTION ID

S49-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles x and 2x are complementary. x =

A. 90°

B. 45°

C. 60°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

1975

QUESTION ID

S49-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 30°

GRADING NOTE 3x = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The complement of θ is 90° − θ. If θ = 64°, find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

1976

QUESTION ID

S49-B1-Q4

TYPE

Short answer

CORRECT ANSWER

26°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Complementary angles must be adjacent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

1977

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They need only add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 2

Opposite and adjacent sides depend on the chosen angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), name the side adjacent to ∠B.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5

1979

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-B2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5

True or false?

In right triangle ABC with the right angle at C, the side opposite ∠A is the side adjacent to ∠B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

1980

QUESTION ID

S49-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both are BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with the right angle at C, cos B equals

A. AC/AB

B. BC/AB

C. AC/BC

D. BC/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

1981

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. BC/AB

GRADING NOTE Adjacent over hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

Write sin A using the sides of right triangle ABC (right angle at C).

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5

1982

QUESTION ID

S49-B2-Q4

TYPE

Short answer

CORRECT ANSWER

BC/AB

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-B2-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin A and cos A are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

1983

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when A = 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 3

Angle sum in a right triangle • Grade 8 • Geometry • 8.G.A.5

A right triangle has an acute angle of 28°. Find the other acute angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

1985

QUESTION ID

S49-B3-Q1

TYPE

Short answer

CORRECT ANSWER

62°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A right triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5

1986

QUESTION ID

S49-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum would exceed 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle, the acute angles are x and x + 20. x =

A. 70°

B. 45°

C. 55°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

1987

QUESTION ID

S49-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 35°

GRADING NOTE 2x + 20 = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do the acute angles of a right triangle add to 90°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

1988

QUESTION ID

S49-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The angles add to 180° and the right angle uses 90°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The two acute angles of an isosceles right triangle are 45° each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

1989

QUESTION ID

S49-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal and complementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 1

Relationships between tangents of complementary angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\tan\theta \cdot \tan(90° - \theta) = 1$ for an acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

1991

GRADING NOTE Any equivalent explanation.

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

tan θ = opposite/adjacent and tan(90° − θ) = adjacent/opposite, which are reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

tan 20° ≈ 0.364. Use the complementary relationship to estimate tan 70° to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

1992

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 2.75

(1 ÷ 0.364 ≈ 2.747)

GRADING NOTE Accept 2.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin \theta = \cos \theta$ exactly when $\theta = 45°$ (for acute θ).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

1993

QUESTION ID

S49-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then $90° - \theta = \theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equals 1 for every acute θ?

A. $\tan\theta + \tan(90° - \theta)$

B. $\tan\theta \cdot \tan(90° - \theta)$

C. $\sin\theta \cdot \cos\theta$

D. $\sin\theta + \cos\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

1994

QUESTION ID

S49-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $\tan\theta \cdot \tan(90° - \theta)$

GRADING NOTE Reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that sin θ / cos(90° − θ) = 1 for acute θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

1995

QUESTION ID

S49-F1-Q5

TYPE

Short answer

CORRECT ANSWER

cos(90° − θ) = sin θ, so the quotient is sin θ / sin θ = 1.

GRADING NOTE Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 2

The Pythagorean identity • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Prove that $(\sin \theta)^2 + (\cos \theta)^2 = 1$ for every acute angle θ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

1997

QUESTION ID

S49-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$(\sin \theta)^2 + (\cos \theta)^2 = 1$

GRADING NOTE REQUIRED WORK: Use a right triangle with legs a, b and hypotenuse c and the Pythagorean Theorem.

Any valid proof (paragraph, two-column, or flow). Essential: $\sin \theta = a/c$, $\cos \theta = b/c$; $a^2 + b^2 = c^2$; divide by c^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 0.28 for acute θ. Find cos θ.

cos θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5

1998

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.96

GRADING NOTE Any equivalent form. √(1 − 0.0784).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin 35° + \cos 35° = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

1999

QUESTION ID

S49-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The squares add to 1; $\sin 35° + \cos 35° \approx 1.39$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = 12/13 (θ acute). sin θ =

A. 1/13

B. 5/13

C. 12/5

D. 13/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5

2000

QUESTION ID

S49-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5/13

GRADING NOTE √(1 − 144/169).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use complementary angles to show $(\sin 20°)^2 + (\sin 70°)^2 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5

2001

QUESTION ID

S49-F2-Q5

TYPE

Short answer

CORRECT ANSWER

sin 70° = cos 20°, so the sum is $(\sin 20°)^2 + (\cos 20°)^2 = 1$.

GRADING NOTE Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q5**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5

SET 50 • HSG-SRT.C.8 • SIMILARITY, RIGHT TRIANGLES, AND TRIGONOMETRY

Find unknown sides of right triangles with trigonometric ratios in applied problems

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Choose the trigonometric ratio that links the known and unknown.	**HSG-SRT.C.6**
BACKWARD 2	Solve one-variable equations with a multiplication or division	**HSA-REI.B.3**
BACKWARD 3	Round to a stated precision and keep track of units	**HSN-Q.A.3**
FORWARD 1	Find a height from two angle-of-elevation observations	**HSG-SRT.C.8**
FORWARD 2	Find a feasible location under a clearance constraint	**HSG-MG.A.3**

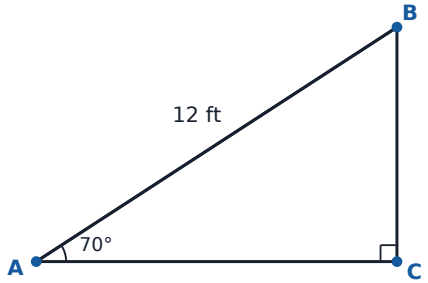
SET 50 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A 12 ft ladder leans against a wall, making a 70° angle with the level ground. How high up the wall does it reach, to the nearest tenth of a foot?



Not drawn to scale.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

2004

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

11.3 ft

GRADING NOTE Rounding to 0.1 required. 12 sin 70° = 11.2763.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A wheelchair ramp rises 2.5 ft and makes an 8° angle with the ground. How long is the ramp surface, to the nearest tenth of a foot?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

2005

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

18.0 ft

GRADING NOTE Rounding to 0.1 required. 2.5 ÷ sin 8° = 17.9632.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

From a point 25 m from the base of a tree, on level ground, the angle of elevation to the top is 38°. How tall is the tree, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

2006

QUESTION ID

S50-M-Q3

TYPE

Short answer

CORRECT ANSWER

19.5 m

GRADING NOTE Rounding to 0.1 required. 25 tan 38° = 19.5321.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A kite string 60 m long makes a 50° angle with the level ground. About how high is the kite above the point where the string is held (nearest meter)?

A. 78 m

B. 46 m

C. 39 m

D. 72 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

2007

QUESTION ID

S50-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 46 m

GRADING NOTE Opposite = hypotenuse × sin 50° = 45.96.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 20 and a 32° angle. Find the leg adjacent to the 32° angle, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

2008

QUESTION ID
S50-M-Q5

TYPE
Short answer

CORRECT ANSWER
17.0

GRADING NOTE Rounding to 0.1 required. 20 cos 32° = 16.9610.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 1

Choose the trigonometric ratio that links the known and unknown sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

You know the hypotenuse and an angle and want the opposite side. Which ratio?

A. none

B. sine

C. cosine

D. tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

2010

QUESTION ID

S50-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. sine

GRADING NOTE Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

You know the adjacent side and an angle and want the opposite side. Which ratio?

A. none

B. tangent

C. cosine

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

2011

QUESTION ID

S50-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. tangent

GRADING NOTE Opposite/adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

2011

ANSWER KEY Reduced copy of the question page, with its answer

Which ratio uses the adjacent side and the hypotenuse?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

2012

QUESTION ID

S50-B1-Q3

TYPE

Short answer

CORRECT ANSWER

Cosine

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

True or false?

To find the hypotenuse from the opposite side and the angle, use sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

2013

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE sin A = opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Tangent uses the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

2014

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Tangent uses the two legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 2

Solve one-variable equations with a multiplication or division • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve x/12 = 0.94.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5

2016

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

11.28

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 0.14 = 2.5/x. Round to the nearest hundredth.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5

2017

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

17.86

GRADING NOTE Rounding to 0.01 required. 2.5 ÷ 0.14 = 17.857....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If sin 30° = h/10, then h =

A. 10 sin 30°

B. 10 ÷ sin 30°

C. sin 30° ÷ 10

D. 30 ÷ 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

2018

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10 sin 30°

GRADING NOTE Multiply both sides by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From 0.5 = 4/x we get x = 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

2019

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE x = 4 ÷ 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From tan 40° = 20/x we get x = 20 tan 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

2020

QUESTION ID
S50-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE x = 20 ÷ tan 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 3

Round to a stated precision and keep track of units • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 11.2763 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5

2022

QUESTION ID

S50-B3-Q1

TYPE

Short answer

CORRECT ANSWER

11.3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B3-Q1**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A calculation gives 19.532 m. Report it to the nearest tenth of a meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

2023

QUESTION ID

S50-B3-Q2

TYPE

Short answer

CORRECT ANSWER

19.5 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B3-Q2**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

2023

True or false?

Rounding in the middle of a calculation can change the final answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

2024

QUESTION ID
S50-B3-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Round at the end.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

2024

ANSWER KEY Reduced copy of the question page, with its answer

A height is measured in feet and the angle in degrees. The answer's unit is

A. degrees

B. feet per degree

C. feet

D. no unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

2025

QUESTION ID
S50-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. feet

GRADING NOTE Lengths keep their units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

17.96 rounded to the nearest tenth is 17.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

2026

QUESTION ID

S50-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 18.0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 1

Find a height from two angle-of-elevation observations • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. Find the tower's height to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5

2028

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

42.0 m

GRADING NOTE Rounding to 0.1 required. h = 40 ÷ (1/tan 25° − 1/tan 40°) = 41.984. Two equations: h = x tan 40° and h = (x + 40) tan 25°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. How far is point B from the base of the tower, to the nearest tenth of a meter?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

2029

QUESTION ID
S50-F1-Q2

TYPE
Short answer

CORRECT ANSWER
50.0 m

GRADING NOTE Rounding to 0.1 required. x = h ÷ tan 40° = 50.034.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

True or false?

With only one angle of elevation and no distance, the height cannot be found.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

2030

QUESTION ID

S50-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One equation has two unknowns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equations describe two observations 40 m apart (angles 25° and 40°, distance x from the closer point)?

A. h = 40 tan 25°

B. h = x tan 25° and h = (x + 40) tan 40°

C. h = x tan 40° and h = (x + 40) tan 25°

D. h = x sin 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

2031

QUESTION ID

S50-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. h = x tan 40° and h = (x + 40) tan 25°

GRADING NOTE The closer point has the larger angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the closer observation point has the larger angle of elevation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5

2032

QUESTION ID

S50-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The same height over a shorter horizontal distance gives a larger tangent, so a larger angle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 2

Find a feasible location under a clearance constraint • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A ramp must rise 3 ft, and its angle with the ground may be at most 4.8°. What is the shortest horizontal run, in whole feet, that meets the requirement?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

2034

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

36 ft

GRADING NOTE Form required: a whole number of feet that meets the constraint (round up). 3 ÷ tan 4.8° = 35.73.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A camera on a 6 m pole must see a point on the ground at an angle of depression of at least 20°. What is the greatest allowed distance from the pole, in tenths of a meter, that still meets the requirement?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

2035

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16.4 m

GRADING NOTE Form required: a distance in tenths of a meter that meets the constraint (round DOWN). 6 ÷ tan 20° = 16.485; farther points need smaller angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

2035

True or false?

Rounding UP the run of a ramp with a maximum angle can make the ramp too steep.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

2036

QUESTION ID

S50-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A longer run makes the ramp less steep; rounding the run down is the risk.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A ladder must reach 10 ft up and make an angle of at most 75° with the ground. To the tenth of a foot, rounded so that the ladder still works, the shortest ladder is

A. 10.0 ft

B. 9.7 ft

C. 2.7 ft

D. 10.4 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

2037

QUESTION ID

S50-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 10.4 ft

GRADING NOTE At exactly 75°, length = 10 ÷ sin 75° = 10.35; a smaller angle needs a longer ladder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A building code allows stairs with an angle of at most 37°. A staircase must rise 9 ft. What is the shortest horizontal run, in tenths of a foot, that meets the code?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

2038

QUESTION ID

S50-F2-Q5

TYPE

Short answer

CORRECT ANSWER

12.0 ft

GRADING NOTE Form required: a run in tenths of a foot that meets the constraint (round UP; 11.9 ft would be too steep). 9 ÷ tan 37° = 11.943.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

SET 51 • HSG-SRT.C.8 • SIMILARITY, RIGHT TRIANGLES, AND TRIGONOMETRY

Find unknown angles with inverse trigonometric ratios in applied problems

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Inverse trig undoes a trig ratio	**HSG-SRT.C.6**
BACKWARD 2	Choose the ratio from the two known sides	**HSG-SRT.C.6**
BACKWARD 3	Use degree mode and round angles	**HSN-Q.A.3**
FORWARD 1	Solve a right triangle completely from two sides	**HSG-SRT.C.8**
FORWARD 2	Find directions (bearings) with inverse trigonometry	**HSG-SRT.C.8**

SET 51 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A ramp rises 1 m over a horizontal distance of 12 m. What angle does it make with the ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

2041

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.8°

GRADING NOTE Rounding to 0.1° required. $\tan^{-1}(1/12) = 4.764°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 6 m ladder reaches 5.5 m up a vertical wall. What angle does it make with the level ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

2042

QUESTION ID
S51-M-Q2

TYPE
Short answer

CORRECT ANSWER
66.4°

GRADING NOTE Rounding to 0.1° required. $\sin^{-1}(5.5/6) = 66.444°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 10 m flagpole casts a 14 m shadow on level ground. Find the angle of elevation of the sun, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

2043

QUESTION ID
S51-M-Q3

TYPE
Short answer

CORRECT ANSWER
35.5°

GRADING NOTE Rounding to 0.1° required. $\tan^{-1}(10/14) = 35.538°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 7 and 24. What is its smaller acute angle, to the nearest tenth of a degree?

A. 73.7°

B. 16.3°

C. 32.5°

D. 26.3°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

2044

QUESTION ID

S51-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 16.3°

GRADING NOTE $\tan^{-1}(7/24) = 16.260°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A road rises 6 m for every 100 m of horizontal distance (a 6% grade). What angle does it make with the horizontal, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

2045

QUESTION ID

S51-M-Q5

TYPE

Short answer

CORRECT ANSWER

3.4°

GRADING NOTE Rounding to 0.1° required. $\tan^{-1}(0.06) = 3.434°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 1

Inverse trig undoes a trig ratio • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 0.5 for an acute angle θ. Find θ.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

2047

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

tan θ = 1 for an acute angle θ. Find θ.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

2048

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which gives an angle from a ratio?

A. cos

B. tan

C. sin

D. $\sin^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

2049

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sin^{-1}$

GRADING NOTE Inverse functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos^{-1}(0.5) = 60°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

2050

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE cos 60° = 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^{-1}(x)$ means $1 \div \sin(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

2051

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the angle whose sine is x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 2

Choose the ratio from the two known sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

You know the opposite side and the hypotenuse. Which inverse function finds the angle?

A. none

B. $\cos^{-1}$

C. $\sin^{-1}$

D. $\tan^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

2053

QUESTION ID
S51-B2-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. $\sin^{-1}$

GRADING NOTE Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

You know both legs. Which inverse function finds an acute angle?

A. $\sin^{-1}$

B. $\tan^{-1}$

C. $\cos^{-1}$

D. none

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

2054

QUESTION ID
S51-B2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. $\tan^{-1}$

GRADING NOTE Opposite/adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rise 5 and run 12. Write the ratio for the angle with the ground.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5

2055

QUESTION ID

S51-B2-Q3

TYPE

Short answer

CORRECT ANSWER

tan θ = 5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5

True or false?

Adjacent side and hypotenuse give the angle by $\cos^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

2056

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cosine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

2056

True or false?

Opposite side and hypotenuse give the angle by $\tan^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

2057

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sin^{-1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 3

Use degree mode and round angles • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 4.7636° to the nearest tenth of a degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5

2059

QUESTION ID

S51-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4.8°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B3-Q1**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5

True or false?

A calculator in radian mode gives $\sin^{-1}(0.5) \approx 0.524$, not 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

2060

QUESTION ID
S51-B3-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An acute angle of a right triangle must be

A. more than 90°

B. any angle

C. exactly 45°

D. between 0° and 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

2061

QUESTION ID

S51-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. between 0° and 90°

GRADING NOTE Acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

2061

ANSWER KEY Reduced copy of the question page, with its answer

A computed angle is 112°. Can it be an acute angle of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

2062

QUESTION ID
S51-B3-Q4

TYPE
Short answer

CORRECT ANSWER
No

GRADING NOTE Any equivalent answer; check the calculation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B3-Q4**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

35.537° to the nearest tenth is 35.6°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

2063

QUESTION ID

S51-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 35.5°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 1

Solve a right triangle completely from two sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 9 and 14. Find the hypotenuse and both acute angles (nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

2065

QUESTION ID
S51-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Hypotenuse 16.6; angles 32.7° and 57.3°

GRADING NOTE Rounding to 0.1 required for all three; the angles add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 17 and one leg 8. Find the other leg and both acute angles (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

2066

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Other leg 15; angles 28.1° and 61.9°

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Once one acute angle is known, the other is 90° minus it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

2067

QUESTION ID

S51-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

Legs 5 and 5. Angles?

A. 45° and 45°

B. 50° and 40°

C. 30° and 60°

D. 5° and 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5

2068

QUESTION ID

S51-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 45° and 45°

GRADING NOTE Isosceles right triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to check a solved right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

2069

QUESTION ID

S51-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The acute angles add to 90°, $a^2 + b^2 = c^2$, and each ratio matches its angle.

GRADING NOTE Any reasonable checks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 2

Find directions (bearings) with inverse trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A boat travels 8 km east and then 5 km north. Find the angle north of east of its straight-line direction from the start, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

2071

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

32.0°

GRADING NOTE Rounding to 0.1° required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A hiker walks 3 km north and 4 km west. What is the bearing angle west of north from the start, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

2072

QUESTION ID
S51-F2-Q2

TYPE
Short answer

CORRECT ANSWER
53.1°

GRADING NOTE Rounding to 0.1° required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Bearing problems become right-triangle problems when the motions are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

2073

QUESTION ID

S51-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE East-west and north-south legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A drone flies 120 m horizontally and rises 50 m. Its angle of climb is about

A. 22.6°

B. 24.6°

C. 67.4°

D. 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

2074

QUESTION ID

S51-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 22.6°

GRADING NOTE $\tan^{-1}(50/120)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A ship is 10 km east and 10 km south of port. Find the straight-line distance (to the nearest tenth of a km) and the angle south of east.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

2075

QUESTION ID

S51-F2-Q5

TYPE

Short answer

CORRECT ANSWER

14.1 km; 45°

GRADING NOTE Form required: both. $10\sqrt{2} \approx 14.1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

Solve right triangles in applied problems with trigonometry and the Pythagorean Theorem

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Use the Pythagorean Theorem to find a side	**8.G.B.7**
BACKWARD 2	Interpret angles of elevation and depression	**HSG-SRT.C.6**
BACKWARD 3	Choose sine, cosine, or tangent	**HSG-SRT.C.6**
FORWARD 1	Solve problems that need two right triangles	**HSG-SRT.C.8**
FORWARD 2	Solve non-right triangles with the Law of Sines	**HSG-SRT.D.11**

SET 52 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From the top of an 80 m cliff, the angle of depression to a boat is 25°. How far is the boat from the base of the cliff, to the nearest tenth of a meter? How far is it from the observer?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

2078

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

171.6 m from the base; 189.3 m from the observer

GRADING NOTE Rounding to 0.1 required for both. Horizontal = 80 ÷ tan 25°; line of sight = 80 ÷ sin 25°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane flying at an altitude of 2,000 m is seen at an angle of elevation of 14°. Find the horizontal ground distance and the line-of-sight distance to the plane, to the nearest meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

2079

QUESTION ID
S52-M-Q2

TYPE
Short answer

CORRECT ANSWER
8022 m horizontal; 8267 m line of sight

GRADING NOTE Rounding to the nearest meter required for both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A ramp is 15 ft long and rises 2 ft. Find its horizontal run (nearest tenth of a foot) and its angle with the ground (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

2080

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

Run 14.9 ft; angle 7.7°

GRADING NOTE Rounding required as stated. Run = $\sqrt{(15^2 - 2^2)}$; angle = $\sin^{-1}(2/15)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

A 25 ft guy wire from the top of a pole meets the level ground 9 ft from the pole. How tall is the pole and what angle does the wire make with the ground (nearest tenth)?

A. 26.6 ft; 68.9°

B. 16.0 ft; 21.1°

C. 23.3 ft; 21.1°

D. 23.3 ft; 68.9°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

2081

QUESTION ID

S52-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 23.3 ft; 68.9°

GRADING NOTE Height $\sqrt{(25^2 - 9^2)}$; angle $\cos^{-1}(9/25)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 25 and a 40° angle. Find both legs and the other acute angle (legs to the nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

2082

QUESTION ID
S52-M-Q5

TYPE
Short answer

CORRECT ANSWER
Legs 16.1 and 19.2; other angle 50°

GRADING NOTE Form required: both legs and the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 1

Use the Pythagorean Theorem to find a side • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 25 and a leg 9. Find the other leg, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

2084

QUESTION ID
S52-B1-Q1

TYPE
Short answer

CORRECT ANSWER
23.3

GRADING NOTE Any equivalent form. √(625 − 81).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

2084

ANSWER KEY Reduced copy of the question page, with its answer

Legs 8 and 15. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

2085

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

17

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

True or false?

In a right triangle, the hypotenuse is the longest side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

2086

QUESTION ID

S52-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is opposite the largest angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 13, leg 5. Other leg?

A. √194

B. 12

C. 8

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

2087

QUESTION ID

S52-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 12

GRADING NOTE √(169 − 25).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a^2 + b^2 = c^2$ works for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q5/5

2088

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 2

Interpret angles of elevation and depression • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

True or false?

An angle of depression is measured down from the horizontal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

2090

QUESTION ID
S52-B2-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The angle of depression from a cliff top to a boat is 25°. What is the angle of elevation from the boat to the cliff top?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

2091

QUESTION ID
S52-B2-Q2

TYPE
Short answer

CORRECT ANSWER
25°

GRADING NOTE Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

Which side is horizontal in an angle-of-elevation triangle?

A. The hypotenuse

B. The leg along the ground

C. The vertical leg

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

2092

QUESTION ID

S52-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The leg along the ground

GRADING NOTE Level ground.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An observer looks up at 30° to a bird. Is that an angle of elevation or depression?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

2093

QUESTION ID

S52-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Elevation

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

True or false?

The angle of depression is measured from the vertical line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

2094

QUESTION ID
S52-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE From the horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 3

Choose sine, cosine, or tangent • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Known: opposite side and angle. Wanted: hypotenuse. Ratio?

A. none

B. tangent

C. sine

D. cosine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

2096

QUESTION ID

S52-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. sine

GRADING NOTE sin = opp/hyp.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

Known: adjacent side and angle. Wanted: opposite side. Ratio?

A. cosine

B. tangent

C. none

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

2097

QUESTION ID

S52-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. tangent

GRADING NOTE tan = opp/adj.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Known: hypotenuse and angle. Wanted: adjacent side. Ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

2098

QUESTION ID

S52-B3-Q3

TYPE

Short answer

CORRECT ANSWER

Cosine

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

True or false?

Two different ratios can lead to the same correct side if used correctly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

2099

QUESTION ID

S52-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, finding the third side by trig or Pythagoras.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Tangent relates the hypotenuse and a leg.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

2100

QUESTION ID

S52-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Tangent relates the two legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 1

Solve problems that need two right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From a window 20 m up, the angle of elevation to the top of a building across the street is 30° and the angle of depression to its base is 40°. How tall is the building, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

2102

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

33.8 m

GRADING NOTE Rounding to 0.1 required. Street width = 20 ÷ tan 40° = 23.835; height = 20 + width · tan 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two observers 100 m apart on opposite sides of a tower see its top at 35° and 50°. Find the tower's height, to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

2103

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

44.1 m

GRADING NOTE Rounding to 0.1 required. h/tan 35° + h/tan 50° = 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

True or false?

A shared side links two right triangles in such problems.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

2104

QUESTION ID

S52-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, a common height or distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two observers 100 m apart on opposite sides of a tower of height h see its top at angles of elevation 35° and 50°. Which equation is correct?

A. h = 100 tan 85°

B. h/sin 35° + h/sin 50° = 100

C. h tan 35° + h tan 50° = 100

D. h/tan 35° + h/tan 50° = 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

2105

QUESTION ID
S52-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. h/tan 35° + h/tan 50° = 100

GRADING NOTE Horizontal distances add to 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two observers on opposite sides of a tower each measure an angle of elevation to its top, and the distance between them is known. Explain why the tower's height cannot be found from one of the two right triangles alone.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

2106

QUESTION ID

S52-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Each triangle alone has two unknowns (h and its distance); the shared height and total distance give the needed second equation.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 2

Solve non-right triangles with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, A = 40°, B = 65°, and a = 12. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5

2108

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

16.9

GRADING NOTE Rounding to 0.1 required. b = a sin B ÷ sin A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, A = 52°, C = 70°, and c = 20. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5

2109

QUESTION ID

S52-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16.8

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5

True or false?

SOH-CAH-TOA applies directly to a triangle with no right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5

2110

QUESTION ID

S52-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Those ratios are defined in right triangles; use the Law of Sines or Cosines, or draw an altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5

2110

Which data allow the Law of Sines directly?

A. Two sides and the included angle

B. Two angles and any side

C. Three sides

D. Three angles only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

2111

QUESTION ID

S52-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Two angles and any side

GRADING NOTE AAS or ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

Explain how dropping an altitude turns a non-right triangle into two right triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5

2112

QUESTION ID
S52-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The altitude creates two right triangles that share it, so $h = b \sin A = a \sin B$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSG-C.A.1 • CIRCLES

Prove that all circles are similar

MAIN	Geometry main question	**HSG-C.A.1**
BACKWARD 1	A dilation multiplies distances by the scale factor	**HSG-SRT.A.1.b**
BACKWARD 2	Translate a point to another point	**8.G.A.3**
BACKWARD 3	A circle is all points at one distance from its center	**HSG-CO.A.1**
FORWARD 1	Find a single dilation that maps one circle onto another	**HSG-C.A.1**
FORWARD 2	Arcs of similar circles scale with the radius	**HSG-C.B.5**

SET 53 • HSG-C.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Circle C has center A and radius r, and circle D has center B and radius s. Prove that the two circles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

2115

QUESTION ID

S53-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation followed by a dilation maps C onto D, so they are similar.

GRADING NOTE REQUIRED WORK: Translate A to B, then dilate about B with scale factor s/r. Show the image of C is exactly D.

Any valid proof (paragraph, two-column, or flow). Essential: (1) the translation maps C onto the circle with center B and radius r (distances kept); (2) the dilation about B with factor s/r maps each point at distance r from B to a point at distance s on the same ray, so onto D; (3) every point of D is reached (the inverse dilation); a similarity transformation exists. Invalid: "circles all look alike"; only checking two specific circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

Circle C_1 has center (1, 2) and radius 3. Circle C_2 has center (−4, 0) and radius 6. Describe a similarity transformation that maps C_1 onto C_2 and explain why it works.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

2116

QUESTION ID

S53-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Translate 5 units left and 2 units down, then dilate about (−4, 0) with scale factor 2.

GRADING NOTE

REQUIRED WORK: Give the translation that moves the centers together and the dilation (center and factor).

Any valid proof (paragraph, two-column, or flow). Essential: the translation (x, y) → (x − 5, y − 2) moves the center to (−4, 0) keeping the radius 3; the dilation about (−4, 0) by 2 doubles every distance from the center, so radius 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle C_1 has center (0, 0) and radius 2. Circle C_2 has center (3, −1) and radius 5. Prove that C_1 and C_2 are similar by giving a similarity transformation, and show where it sends the point (2, 0) of C_1.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

2117

QUESTION ID

S53-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Translate by (x, y) → (x + 3, y − 1), then dilate about (3, −1) by 2.5; (2, 0) → (5, −1) → (8, −1), which is 5 from (3, −1).

GRADING NOTE REQUIRED WORK: Translate the center onto (3, −1), then dilate about (3, −1) with factor 5/2; follow (2, 0) through both steps and check its distance from (3, −1).

Any valid proof (paragraph, two-column, or flow). Essential: the translation keeps the radius 2 and moves the center to (3, −1); the dilation by 5/2 about the center makes every radius 5, so the image is C_2; the check point lands on C_2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why two circles with the same center but different radii, r and s, are similar using only ONE transformation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

2118

QUESTION ID

S53-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The dilation about the common center with factor s/r maps the first circle onto the second.

GRADING NOTE REQUIRED WORK: Name the transformation and its center and factor.

Any valid proof (paragraph, two-column, or flow). Essential: dilation about the shared center; factor s/r; each point at distance r goes to distance s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle C has center A and radius r, and circle D has center B and radius s. Prove that C and D are similar using a dilation FIRST and a translation SECOND.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

2119

QUESTION ID

S53-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The dilation followed by the translation maps C onto D, so they are similar.

GRADING NOTE REQUIRED WORK: Dilate about A with factor s/r, then translate A to B; show the image is exactly D.

Any valid proof (paragraph, two-column, or flow). Essential: the dilation about A by s/r maps C onto the circle with center A and radius s (every distance from A is multiplied by s/r); the translation A → B keeps distances, so it maps that circle onto the circle with center B and radius s, which is D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 1

A dilation multiplies distances by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

A dilation with factor 2.5 is applied to a segment of length 4. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

2121

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A dilation about point O keeps O fixed and multiplies each distance from O by the factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5

2122

QUESTION ID
S53-B1-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5

Which factor maps a circle of radius 4 about its center to one of radius 10?

A. 2.5

B. 0.4

C. 6

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

2123

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.5

GRADING NOTE 10 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius 9 is dilated about its center by ⅓. New radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

2124

QUESTION ID
S53-B1-Q4

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A dilation can turn a circle into a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

2125

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It keeps shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 2

Translate a point to another point • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which translation moves (1, 2) onto (−4, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

2127

QUESTION ID

S53-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5 units left and 2 units down

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

True or false?

A translation keeps the radius of a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5

2128

QUESTION ID

S53-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5

A circle centered at (3, 3) is translated by (x, y) → (x − 3, y + 1). New center?

A. (6, 2)

B. (0, 4)

C. (0, 3)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

2129

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (0, 4)

GRADING NOTE Apply the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A translation moves (2, −1) to (7, 4). Describe it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

2130

QUESTION ID

S53-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5 units right and 5 units up

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A translation changes the size of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5

2131

QUESTION ID

S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 3

A circle is all points at one distance from its center • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every point of a circle is the same distance from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

2133

QUESTION ID

S53-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (0, 0) and passes through (3, 4). Its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

2134

QUESTION ID

S53-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is completely determined by

A. its color

B. one point on it

C. its center only

D. its center and radius

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

2135

QUESTION ID

S53-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. its center and radius

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two circles have the same center and the same radius. What can you say?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

2136

QUESTION ID

S53-B3-Q4

TYPE

Short answer

CORRECT ANSWER

They are the same circle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

True or false?

A circle's radius can be different in different directions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

2137

QUESTION ID

S53-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The definition requires one distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSG-C.A.1

FORWARD 1

Find a single dilation that maps one circle onto another • High School — Geometry • Geometry: Circles • HSG-C.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Circle P: center (0, 0), radius 2. Circle Q: center (6, 0), radius 4. Find the center of the dilation with a positive factor that maps P onto Q.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

2139

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(−6, 0)

GRADING NOTE Form required: an ordered pair. The center (c, 0) satisfies 6 = c + 2(0 − c), so c = −6; check: (0, 0) → −6 + 2(0 + 6) = 6, and the radius 2 → 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

Circle P has center (0, 0) and radius 2. Circle Q has center (6, 0) and radius 4. What is the scale factor of a dilation that maps P onto Q?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

2140

QUESTION ID
S53-F1-Q2

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form. 4 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

True or false?

Two circles with different radii can always be mapped onto each other by one dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5

2141

QUESTION ID

S53-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Choose the center on the line of centers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5

Circles with equal radii and different centers are related by

A. nothing

B. a translation (no dilation needed)

C. a dilation with factor 2

D. a reflection only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

2142

QUESTION ID

S53-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a translation (no dilation needed)

GRADING NOTE Same size: congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

Circle A has radius 3 and circle B has radius 12, both centered at the origin. Which dilation maps A onto B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

2143

QUESTION ID

S53-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Center (0, 0), factor 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSG-C.A.1

FORWARD 2

Arcs of similar circles scale with the radius • High School — Geometry • Geometry: Circles • HSG-C.B.5

Two circles have radii 3 and 7. A 60° central angle cuts off an arc in each. What is the ratio of the arc lengths?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

2145

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3 : 7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

A 90° arc of a circle with radius 4 has length 2π. How long is a 90° arc of a circle with radius 10?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

2146

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

True or false?

For a fixed central angle, the arc length is the same in every circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

2147

QUESTION ID

S53-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Arc length is proportional to the radius; s/r is what stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc of length 6 in a circle of radius 2 corresponds, by similarity, to what arc length in a circle of radius 5 (same angle)?

A. 2.4

B. 15

C. 30

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

2148

QUESTION ID

S53-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 15

GRADING NOTE Multiply by 5/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

Explain how the similarity of circles leads to radian measure.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

2149

QUESTION ID

S53-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Since arc length is proportional to radius for a fixed angle, arc/radius depends only on the angle; that number is the angle in radians.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

SET 54 • HSG-C.A.2 • CIRCLES

Relate inscribed angles and central angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Base angles of an isosceles triangle are congruent	**HSG-CO.C.10**
BACKWARD 2	The exterior angle theorem	**HSG-CO.C.10**
BACKWARD 3	A central angle and its arc have the same measure	**4.MD.C.5.a**
FORWARD 1	Prove the inscribed angle theorem when the center is inside or out…	**HSG-C.A.2**
FORWARD 2	Inscribed angles on the same arc, and arc applications	**HSG-C.A.2**

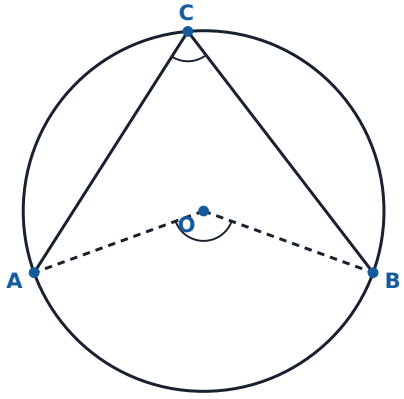
SET 54 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, central angle AOB measures 110°. Inscribed angle ACB intercepts the same arc AB. Find m∠ACB.



m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

2152

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

55°

GRADING NOTE Any equivalent form. An inscribed angle is half the central angle on the same arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, inscribed angle PQR measures 38°. Find the measure of arc PR (the arc it intercepts) and of central angle POR.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

2153

QUESTION ID
S54-M-Q2

TYPE
Short answer

CORRECT ANSWER
Arc PR = 76°;
∠POR = 76°

GRADING NOTE Form required: both. The central angle equals its arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, AC is a diameter and B is another point on the circle. Prove that m∠BAC = ½ m∠BOC (the inscribed angle theorem when one side of the angle is a diameter).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

2154

QUESTION ID

S54-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠BAC = ½ m∠BOC

GRADING NOTE REQUIRED WORK: Use isosceles triangle AOB (OA = OB) and the exterior angle theorem at O.

Any valid proof (paragraph, two-column, or flow). Essential: OA = OB (radii), so ∠OAB ≅ ∠OBA; ∠BOC is an exterior angle of △AOB, so m∠BOC = m∠OAB + m∠OBA = 2m∠BAC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed angle XYZ and central angle XOZ intercept the same arc XZ, and m∠XYZ = 47°. What is m∠XOZ?

A. 23.5°

B. 94°

C. 47°

D. 133°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

2155

QUESTION ID
S54-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 94°

GRADING NOTE Double the inscribed angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, points C and D are on the same side of chord AB, so inscribed angles ADB and ACB intercept the same arc AB. m∠ADB = (3x + 4)° and m∠ACB = (5x − 12)°. Find x, m∠ACB, and the central angle AOB that intercepts arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

2156

QUESTION ID

S54-M-Q5

TYPE

Short answer

CORRECT ANSWER

x = 8; m∠ACB = 28°; m∠AOB = 56°.

GRADING NOTE Form required: all three. Both inscribed angles are half of arc AB, so 3x + 4 = 5x − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 1

Base angles of an isosceles triangle are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

In △AOB, OA = OB and m∠OAB = 35°. Find m∠OBA and m∠AOB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

2158

QUESTION ID
S54-B1-Q1

TYPE
Short answer

CORRECT ANSWER
35° and 110°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

True or false?

Two radii of a circle and the chord joining their ends form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

2159

QUESTION ID
S54-B1-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The radii are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Isosceles triangle with vertex angle 80°. Each base angle?

A. 100°

B. 50°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

2160

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 50°

GRADING NOTE (180 − 80)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △OPQ with OP = OQ, m∠P = 2x and m∠Q = 50°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

2161

QUESTION ID

S54-B1-Q4

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

True or false?

In an isosceles triangle, the vertex angle always equals a base angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

2162

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when equilateral.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 2

The exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has remote interior angles 30° and 30°. Find the exterior angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

2164

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

True or false?

An exterior angle equals the sum of the two remote interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

2165

QUESTION ID

S54-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

Exterior angle 100°, one remote interior angle 45°. The other?

A. 45°

B. 80°

C. 55°

D. 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

2166

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 55°

GRADING NOTE 100 − 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If both remote interior angles equal x, the exterior angle is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

2167

QUESTION ID

S54-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

True or false?

An exterior angle equals the adjacent interior angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

2168

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 3

A central angle and its arc have the same measure • Grade 4 • Measurement & Data • 4.MD.C.5.a

ANSWER KEY Reduced copy of the question page, with its answer

A central angle cuts off ¼ of a circle. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

2170

QUESTION ID

S54-B3-Q1

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

True or false?

The measure of a minor arc equals its central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

2171

QUESTION ID

S54-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of arc measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc measures 70°. Its central angle measures

A. 70°

B. 140°

C. 290°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

2172

QUESTION ID

S54-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 70°

GRADING NOTE Equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A central angle of 150°. What fraction of the circle is its arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

2173

QUESTION ID

S54-B3-Q4

TYPE

Short answer

CORRECT ANSWER

5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

True or false?

All arcs of 60° have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

2174

QUESTION ID

S54-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Length depends on the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 1

Prove the inscribed angle theorem when the center is inside or outside the angle • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed angle ABC in circle O has the center O inside the angle. Prove that m∠ABC = ½ m(arc AC), where arc AC is the arc intercepted by ∠ABC (the arc not containing B). (Draw diameter BD.) If ∠ABC is obtuse, arc AC is a major arc and its central angle is a reflex angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

2176

QUESTION ID

S54-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠ABC = ½ m(arc AC)

GRADING NOTE REQUIRED WORK: Split ∠ABC along BD, apply the diameter case to each part, then add the arcs.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ABD = ½ m(arc AD) and m∠DBC = ½ m(arc DC) (diameter case, with each arc measured by its central angle); D lies on arc AC, so m∠ABC = ½(m(arc AD) + m(arc DC)) = ½ m(arc AC). This also covers an obtuse ∠ABC, where arc AC is greater than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed angle ABC in circle O has the center O outside the angle. Prove that m∠ABC = ½ m(arc AC), where arc AC is the arc intercepted by ∠ABC (the arc not containing B). (Draw diameter BD.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

2177

QUESTION ID

S54-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

m∠ABC = ½ m(arc AC)

GRADING NOTE REQUIRED WORK: Use the diameter case twice and SUBTRACT the arcs.

Any valid proof (paragraph, two-column, or flow). Essential: label so that A lies on arc DC (A and C on the same side of BD); m∠DBC = ½ m(arc DC) and m∠DBA = ½ m(arc DA) (diameter case); subtract: m∠ABC = ½(m(arc DC) − m(arc DA)) = ½ m(arc AC). Here ∠ABC is acute and arc AC is a minor arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

True or false?

The inscribed angle theorem needs three cases for a complete proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

2178

QUESTION ID

S54-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Center on a side, inside, or outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

In the proof of the inscribed angle theorem for the case where the center lies inside the inscribed angle, a diameter splits the angle in two, and the two diameter-case results are

A. averaged

B. multiplied

C. subtracted

D. added

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

2179

QUESTION ID

S54-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. added

GRADING NOTE The angle is the sum of two pieces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can't the proof of the inscribed angle theorem stop after the case in which one side of the angle is a diameter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

2180

QUESTION ID

S54-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Most inscribed angles do not have a diameter as a side; the other positions must be covered too.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 2

Inscribed angles on the same arc, and arc applications • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In a circle, inscribed angles ∠ADB and ∠ACB intercept the same arc AB, and m∠ADB = 41°. Find m∠ACB.

m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

2182

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

41°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a circle, inscribed angles ADB and ACB intercept the same arc AB (C and D are on the same side of chord AB). Prove that ∠ADB ≅ ∠ACB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

2183

QUESTION ID

S54-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

They are congruent.

GRADING NOTE REQUIRED WORK: Each inscribed angle is half the measure of the same intercepted arc.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ADB = ½ m(arc AB) and m∠ACB = ½ m(arc AB), where arc AB is the arc not containing C or D (it may be a major arc, with a reflex central angle); equal halves of the same arc, so the angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

True or false?

An inscribed angle whose intercepted arc is a 200° major arc measures 200°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

2184

QUESTION ID

S54-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is half the intercepted arc: 100°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Points A, B, C, D are on a circle with arc AB = 80° and arc CD = 50° (arcs not containing the other points). Chords AC and BD meet at E inside. m∠AEB?

A. 30°

B. 65°

C. 15°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

2185

QUESTION ID

S54-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 65°

GRADING NOTE Angle formed by two chords = half the sum of the intercepted arcs: (80 + 50)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle measures 3x + 10 and its intercepted arc measures 110°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

2186

QUESTION ID

S54-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 3x + 10 = 55.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

2186

SET 55 • HSG-C.A.2 • CIRCLES

Inscribed angles on a diameter are right angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Isosceles triangles from radii	**HSG-CO.C.10**
BACKWARD 3	Triangle angle sum	**8.G.A.5**
FORWARD 1	Use the converse to find a circle through a right triangle	**HSG-C.A.2**
FORWARD 2	Use the diameter property to construct tangents from an external…	**HSG-C.A.4**

SET 55 • HSG-C.A.2

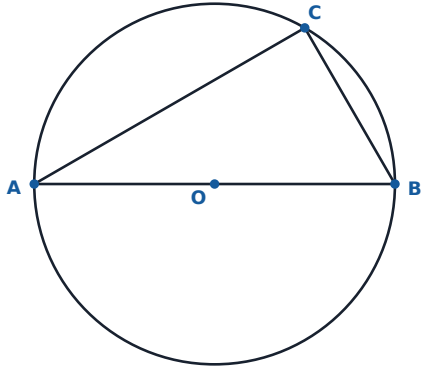
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

AB is a diameter of circle O and C is any other point on the circle. Prove that ∠ACB is a right angle.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5

2189

QUESTION ID

S55-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠ACB = 90°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem (the intercepted arc is 180°), or the two isosceles triangles AOC and BOC.

Any valid proof (paragraph, two-column, or flow). Essential (theorem): ∠ACB intercepts a semicircle, 180°, so it measures 90°. Alternative: OA = OC = OB, so △AOC and △BOC are isosceles; with base angles x and y, the angle sum of △ABC gives 2x + 2y = 180°, so x + y = 90° = m∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5

2189

ANSWER KEY Reduced copy of the question page, with its answer

AB is a diameter of a circle and C is on the circle. AC = 6 and BC = 8. Find AB and the radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

2190

QUESTION ID

S55-M-Q2

TYPE

Short answer

CORRECT ANSWER

AB = 10; radius 5

GRADING NOTE Form required: both. ∠C = 90°, so AB = √(36 + 64).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

AB is a diameter of circle O and C is on the circle with m∠CAB = 34°. Find m∠CBA.

m∠CBA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

2191

QUESTION ID
S55-M-Q3

TYPE
Short answer

CORRECT ANSWER
56°

GRADING NOTE Any equivalent form. ∠C = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

Triangle PQR is inscribed in a circle with PR a diameter. m∠P = (3x)° and m∠R = (2x + 15)°. Find x and the measures of all three angles, and name the theorem you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

2192

QUESTION ID

S55-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 15; m∠P = 45°, m∠R = 45°, m∠Q = 90° (an angle inscribed in a semicircle is a right angle).

GRADING NOTE Form required: x, three angles, and the theorem. 3x + 2x + 15 = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Points A(−5, 0), B(5, 0) and C(3, 4) are on the circle $x^2 + y^2 = 25$, and AB is a diameter. Use slopes to show that ∠ACB is a right angle, and explain how this agrees with the theorem.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

2193

QUESTION ID

S55-M-Q5

TYPE

Short answer

CORRECT ANSWER

Slope CA = 4/8 = ½ and slope CB = 4/(−2) = −2; the product is −1, so CA ⊥ CB. This matches the theorem: ∠ACB is inscribed in a semicircle.

GRADING NOTE Form required: both slopes, the perpendicularity conclusion, and the link to the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 1

Use the inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle intercepts an arc of 140°. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

2195

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

70°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle is half its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

2196

QUESTION ID

S55-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A semicircle measures

A. 360°

B. 180°

C. 90°

D. 120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

2197

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 180°

GRADING NOTE Half the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle measures 25°. Its arc?

Arc: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

2198

QUESTION ID

S55-B1-Q4

TYPE

Short answer

CORRECT ANSWER

50°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

2199

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 2

Isosceles triangles from radii • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

OA = OC (radii) and m∠OAC = 40°. Find m∠OCA.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

2201

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

40°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

True or false?

Any two radii and the chord between them form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

2202

QUESTION ID

S55-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △OBC with OB = OC, m∠BOC = 100°. m∠OBC =

A. 80°

B. 40°

C. 100°

D. 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

2203

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40°

GRADING NOTE (180 − 100)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why are ∠OAC and ∠OCA congruent when OA = OC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

2204

QUESTION ID

S55-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Base angles of an isosceles triangle are congruent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

True or false?

A radius is always longer than a chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

2205

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Chords can be longer (up to the diameter).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 3

Triangle angle sum • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle are 90° and 34°. The third?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

2207

QUESTION ID

S55-B3-Q1

TYPE

Short answer

CORRECT ANSWER

56°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The acute angles of a right triangle add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

2208

QUESTION ID

S55-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 180 − 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2x + 2y = 180$. Then $x + y =$

A. 45

B. 360

C. 90

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

2209

QUESTION ID

S55-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 90

GRADING NOTE Divide by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles x, x, and 2x. Find 2x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

2210

QUESTION ID

S55-B3-Q4

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A triangle can have angles 90°, 60°, and 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

2211

QUESTION ID

S55-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 190°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSG-C.A.2

FORWARD 1

Use the converse to find a circle through a right triangle • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C, with A(0, 0) and B(8, 6). Where is the center of the circle through A, B, and C, and what is its radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

2213

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Center (4, 3), the midpoint of AB; radius 5.

GRADING NOTE Form required: both. The hypotenuse is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, m∠C = 90°. Prove that C lies on the circle with diameter AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

2214

QUESTION ID

S55-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

MC = MA = MB, so C is on the circle with center M and diameter AB.

GRADING NOTE REQUIRED WORK: Let M be the midpoint of AB and show MC = MA (for example, by completing a rectangle).

Any valid proof (paragraph, two-column, or flow). Essential: reflect C through M to D; ACBD is a parallelogram with a right angle, so a rectangle; its diagonals are congruent and bisect each other, so MC = MA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

True or false?

The hypotenuse of a right triangle is a diameter of its circumscribed circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

2215

QUESTION ID

S55-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

A right triangle has legs 5 and 12. Radius of its circumscribed circle?

A. 6.5

B. 8.5

C. 6

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

2216

QUESTION ID

S55-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6.5

GRADING NOTE Half the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

A carpenter's square (a right angle) is placed with its corner touching a circular board and its edges crossing the edge at P and Q. What is segment PQ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

2217

QUESTION ID
S55-F1-Q5

TYPE
Short answer

CORRECT ANSWER
A diameter of the board.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSG-C.A.2

FORWARD 2

Use the diameter property to construct tangents from an external point • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

Compass and straightedge only; leave all arcs.

Construct the two tangent lines from point P to circle O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

2219

QUESTION ID

S55-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the midpoint M of OP; draw the circle with center M through O and P; it meets circle O at T_1 and T_2; lines PT_1 and PT_2 are tangents.

GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular bisector of OP, the circle on diameter OP, and both tangent lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

To construct tangents from P to circle O, a circle with diameter OP is drawn; it meets circle O at T_1 and T_2. Why are lines PT_1 and PT_2 tangent to circle O?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

2220

QUESTION ID
S55-F2-Q2

TYPE
Short answer

CORRECT ANSWER

T_1 is on the circle with diameter OP, so $\angle OT_1P = 90°$; a line perpendicular to a radius at its endpoint is tangent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

True or false?

From a point outside a circle there is exactly one tangent line to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

2221

QUESTION ID

S55-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

To construct a tangent from a point P outside circle O, a circle with diameter OP is drawn. That circle is used because

A. it passes through the center only

B. every angle inscribed in it on OP is a right angle

C. it has the same radius as circle O

D. it is tangent to circle O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

2222

QUESTION ID

S55-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. every angle inscribed in it on OP is a right angle

GRADING NOTE Right angle at the tangent point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

O is the center, OP = 10, and the radius is 6. How long is each tangent segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

2223

QUESTION ID

S55-F2-Q5

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

SET 56 • HSG-C.A.2 • CIRCLES

A radius is perpendicular to the tangent at the point of tangency

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the Pythagorean Theorem and its converse	**8.G.B.7**
BACKWARD 2	The perpendicular is the shortest segment from a point to a line	**7.G.A.2**
BACKWARD 3	A tangent touches a circle at exactly one point	**HSG-CO.A.1**
FORWARD 1	Tangent segments from an external point are congruent	**HSG-C.A.2**
FORWARD 2	Write the equation of a tangent line in the coordinate plane	**HSG-GPE.B.5**

SET 56 • HSG-C.A.2

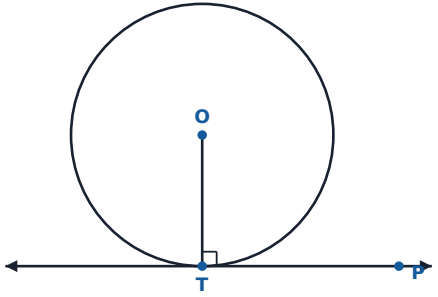
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is tangent to circle O at T. Prove that OT $\perp \ell$. (Use an indirect argument.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5

2226

QUESTION ID

S56-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

OT $\perp \ell$

GRADING NOTE REQUIRED WORK: Suppose OT is not perpendicular; drop the perpendicular from O to ℓ at Q ≠ T; show a second point of ℓ is on the circle or inside it.

Any valid proof (paragraph, two-column, or flow). Essential: if OQ $\perp \ell$ with Q ≠ T, then OQ < OT (the perpendicular is the shortest segment, as the hypotenuse OT is longest in right △OQT), so Q is inside the circle, and then ℓ would cross the circle twice, contradicting tangency. Invalid: "it looks perpendicular".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Line PT is tangent to circle O at T. OT = 5 and PT = 12. Find OP.

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

2227

QUESTION ID
S56-M-Q2

TYPE
Short answer

CORRECT ANSWER
13

GRADING NOTE Any equivalent form. ∠OTP = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

A tangent from P touches circle O at T. OP = 17 and the radius is 8. Find PT.

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

2228

QUESTION ID

S56-M-Q3

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

T lies on circle O, and P is outside it, with OT = 7, PT = 24 and OP = 26. Is line PT tangent to the circle at T? Justify your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

2229

QUESTION ID

S56-M-Q4

TYPE

Short answer

CORRECT ANSWER

No, PT is not tangent: $7^2 + 24^2 = 625 \neq 676 = 26^2$, so $\angle OTP$ is not a right angle, and a tangent at T must be perpendicular to radius OT.

GRADING NOTE Form required: the computation, the conclusion "not tangent", and the reason (radius $\perp$ tangent).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

T is on circle O with radius 6. Line ℓ is perpendicular to OT at T, and Q is a point on ℓ with TQ = 8. Find OQ, and explain why Q is outside the circle. What does this show about line ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

2230

QUESTION ID

S56-M-Q5

TYPE

Short answer

CORRECT ANSWER

OQ = 10 > 6, so Q is outside. Every point of ℓ other than T is the end of a hypotenuse longer than OT, so ℓ meets the circle only at T: ℓ is tangent.

GRADING NOTE Form required: the length, the comparison, and the tangent conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 1

Use the Pythagorean Theorem and its converse • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Legs 5 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

2232

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 17, leg 8. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

2233

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

True or false?

A triangle with sides 7, 24, 25 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

2234

QUESTION ID

S56-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 49 + 576 = 625.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

Sides 6, 8, 11. Right triangle?

A. Yes

B. No

C. Only if 11 is a leg

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

2235

QUESTION ID

S56-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. No

GRADING NOTE 36 + 64 ≠ 121.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The Pythagorean Theorem applies to any triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

2236

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 2

The perpendicular is the shortest segment from a point to a line • Grade 7 • Geometry • 7.G.A.2

True or false?

The shortest segment from a point to a line is perpendicular to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

2238

QUESTION ID

S56-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any other segment is a hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle, which side is longest?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

2239

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The hypotenuse

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

O is 4 cm from line ℓ (perpendicular distance). Another segment from O to ℓ could be

A. 4 cm only

B. 5 cm

C. 2 cm

D. 3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

2240

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5 cm

GRADING NOTE At least 4 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

Why is the perpendicular segment shorter than any slanted one?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

2241

QUESTION ID

S56-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Each slanted segment is the hypotenuse of a right triangle with the perpendicular as a leg.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

True or false?

The distance from a point to a line is measured along any segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

2242

QUESTION ID

S56-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Along the perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

2242

SET 56 • HSG-C.A.2

BACKWARD 3

A tangent touches a circle at exactly one point • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

A tangent line meets the circle at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

2244

QUESTION ID
S56-B3-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

A line meets a circle at two points. It is a

A. secant

B. radius

C. diameter

D. tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

2245

QUESTION ID

S56-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. secant

GRADING NOTE Two points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

Point Q on a line is inside the circle. Can the line be tangent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

2246

QUESTION ID

S56-B3-Q3

TYPE

Short answer

CORRECT ANSWER

No; it would cross the circle twice.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

What is the point where a tangent touches the circle called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

2247

QUESTION ID

S56-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The point of tangency

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

True or false?

A tangent can pass through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

2248

QUESTION ID

S56-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Then it would meet the circle twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

SET 56 • HSG-C.A.2

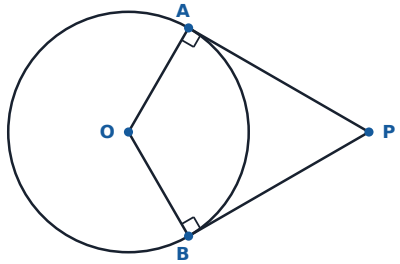
FORWARD 1

Tangent segments from an external point are congruent • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From point P outside circle O, tangent segments PA and PB touch the circle at A and B. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

2250

QUESTION ID

S56-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

PA = PB

GRADING NOTE REQUIRED WORK: Use the right angles at A and B, OA = OB, and the shared hypotenuse OP (HL, or the Pythagorean Theorem).

Any valid proof (paragraph, two-column, or flow). Essential: ∠OAP = ∠OBP = 90°; OA = OB; OP shared; HL (or PA = $\sqrt{(OP^2 - r^2)}$ = PB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangent segments from P to a circle are 3x + 4 and 25. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

2251

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

True or false?

The segment from P to the center bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

2252

QUESTION ID
S56-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

A circle is inscribed in a triangle with sides tangent to it. From one vertex, the two tangent segments are

A. equal

B. parallel

C. unequal

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

2253

QUESTION ID

S56-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. equal

GRADING NOTE Same external point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is tangent to all sides of a quadrilateral ABCD. AB = 9, BC = 11, CD = 13. Find DA.

DA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

2254

QUESTION ID

S56-F1-Q5

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form. AB + CD = BC + DA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSG-C.A.2

FORWARD 2

Write the equation of a tangent line in the coordinate plane • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find an equation of the line tangent to $x^2 + y^2 = 25$ at (3, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5

2256

QUESTION ID

S56-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{3}{4}x + 25/4$

GRADING NOTE Any equivalent form, such as 3x + 4y = 25. The radius has slope 4/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find an equation of the tangent to $x^2 + y^2 = 13$ at $(2, -3)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5

2257

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$2x - 3y = 13$

GRADING NOTE Any equivalent form. Radius slope $-3/2$; tangent slope $2/3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5

True or false?

A tangent line has the same slope as the radius to its point of tangency.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

2258

QUESTION ID

S56-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its slope is the negative reciprocal (when neither line is vertical or horizontal).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The tangent to $x^2 + y^2 = 16$ at (0, 4) is

A. $x = 0$

B. $y = x + 4$

C. $x = 4$

D. $y = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5

2259

QUESTION ID

S56-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. y = 4

GRADING NOTE The radius is vertical, so the tangent is horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is y = 7 tangent to the circle with center (2, 3) and radius 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

2260

GRADING NOTE Any equivalent answer.

QUESTION ID

S56-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: the distance from (2, 3) to the line y = 7 is 4, the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

SET 57 • HSG-C.A.2 • CIRCLES

Relate circumscribed angles and central angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	A tangent is perpendicular to the radius at the point of tangency	**HSG-C.A.2**
BACKWARD 2	The angles of a quadrilateral add to 360°	**8.G.A.5**
BACKWARD 3	Supplementary angles add to 180°	**7.G.B.5**
FORWARD 1	Use the circumscribed angle with arcs, including major arcs	**HSG-C.A.2**
FORWARD 2	Angles formed by secants and tangents outside a circle	**HSG-C.A.2**

SET 57 • HSG-C.A.2

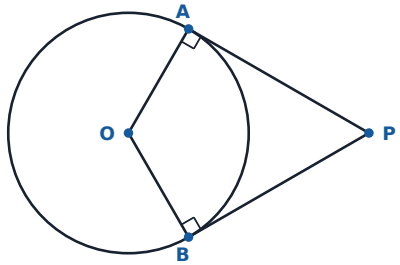
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Prove that m∠APB + m∠AOB = 180°.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

2263

QUESTION ID

S57-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠APB + m∠AOB = 180°

GRADING NOTE REQUIRED WORK: Use the right angles at A and B and the angle sum of quadrilateral OAPB.

Any valid proof (paragraph, two-column, or flow). Essential: ∠OAP = ∠OBP = 90° (radius ⊥ tangent); the angles of quadrilateral OAPB add to 360°; so m∠APB + m∠AOB = 360° − 180° = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. The circumscribed angle APB measures 52°. Find the central angle AOB and the minor arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

2264

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

∠AOB = 128°; arc AB = 128°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Minor arc AB measures 100°. Find the circumscribed angle APB.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

2265

QUESTION ID
S57-M-Q3

TYPE
Short answer

CORRECT ANSWER
80°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. m∠APB = 90°. Find m∠AOB and classify quadrilateral OAPB as specifically as possible. Give reasons.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

2266

QUESTION ID

S57-M-Q4

TYPE

Short answer

CORRECT ANSWER

m∠AOB = 90°. OAPB is a square: all four angles are 90° (a rectangle), and OA = OB (radii) and PA = PB (tangent segments) with OA = PB (opposite sides), so all sides are equal.

GRADING NOTE Form required: the angle, "square", and the reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. m∠APB = (2x + 10)° and m∠AOB = (4x + 50)°. Find x and both angle measures, and state the relationship you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

2267

QUESTION ID
S57-M-Q5

TYPE
Short answer

CORRECT ANSWER
x = 20; m∠APB = 50°, m∠AOB = 130°; a circumscribed angle and its central angle are supplementary.

GRADING NOTE Form required: x, both angles, and the relationship. 6x + 60 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 1

A tangent is perpendicular to the radius at the point of tangency • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

The radius to the point of tangency is perpendicular to the tangent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

2269

QUESTION ID

S57-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is m∠OAP when PA is tangent at A?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

2270

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

In quadrilateral OAPB with tangents PA and PB, how many right angles are certain?

A. 4

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

2271

QUESTION ID

S57-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE At A and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangent PA, radius OA = 9, PA = 12. OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

2272

QUESTION ID

S57-B1-Q4

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The angle between a tangent and a radius can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

2273

QUESTION ID

S57-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Always 90° at the point of tangency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 2

The angles of a quadrilateral add to 360° • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

A quadrilateral has angles 90°, 90°, and 70°. The fourth?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

2275

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

110°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

True or false?

A diagonal splits a quadrilateral into two triangles, so its angles add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

2276

QUESTION ID

S57-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2 × 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a quadrilateral are right angles. The other two must

A. each be 90°

B. add to 360°

C. add to 180°

D. add to 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

2277

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. add to 180°

GRADING NOTE 360 − 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles x, 90°, 90°, 2x in a quadrilateral. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

2278

QUESTION ID

S57-B2-Q4

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

True or false?

The angles of a quadrilateral add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

2279

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 3

Supplementary angles add to 180° • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find the supplement of 52°.

Supplement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

2281

QUESTION ID

S57-B3-Q1

TYPE

Short answer

CORRECT ANSWER

128°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Supplementary angles add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

2282

QUESTION ID

S57-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles x and 3x are supplementary. x =

A. 60°

B. 90°

C. 45°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

2283

QUESTION ID

S57-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 45°

GRADING NOTE 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If two angles are supplementary and one is obtuse, the other is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

2284

QUESTION ID

S57-B3-Q4

TYPE

Short answer

CORRECT ANSWER

acute

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

True or false?

Supplementary angles must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

2285

QUESTION ID

S57-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They need not be adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSG-C.A.2

FORWARD 1

Use the circumscribed angle with arcs, including major arcs • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. The circumscribed angle APB is 40°. Find the minor arc AB and the major arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

2287

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Minor arc 140°; major arc 220°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Show that m∠APB = ½(major arc AB − minor arc AB).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

2288

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

If the minor arc is m, the major arc is 360 − m and the angle is 180 − m = ½((360 − m) − m).

GRADING NOTE Any equivalent algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

True or false?

A circumscribed angle can measure more than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

2289

QUESTION ID

S57-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 180° minus the minor arc, so it is less than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. m∠APB = 70°. An inscribed angle intercepting the minor arc AB measures

A. 110°

B. 70°

C. 55°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

2290

QUESTION ID
S57-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 55°

GRADING NOTE Minor arc = 110°, inscribed angle = 55°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

2290

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Minor arc AB is x°, and the circumscribed angle APB is (x − 20)°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

2291

QUESTION ID

S57-F1-Q5

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form. x + x − 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSG-C.A.2

FORWARD 2

Angles formed by secants and tangents outside a circle • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Two secants from P meet a circle at A, B and at C, D (A between P and B, C between P and D). Prove that m∠P = ½(arc BD − arc AC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

2293

QUESTION ID

S57-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠P = ½(arc BD − arc AC)

GRADING NOTE REQUIRED WORK: Draw chord BC; use the exterior angle theorem in triangle PBC and the inscribed angle theorem.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BCD is exterior to △PBC, so m∠BCD = m∠P + m∠PBC; m∠BCD = ½ arc BD and m∠PBC = ½ arc AC; subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

2293

ANSWER KEY Reduced copy of the question page, with its answer

Two secants from P intercept arcs of 110° and 30°. Find m∠P.

m∠P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

2294

QUESTION ID
S57-F2-Q2

TYPE
Short answer

CORRECT ANSWER
40°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

True or false?

The angle outside a circle formed by a tangent and a secant is half the sum of the intercepted arcs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

2295

QUESTION ID

S57-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is half the difference; half the sum is for two chords meeting inside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

An angle outside a circle intercepts arcs of 160° and 50°. Its measure?

A. 110°

B. 55°

C. 210°

D. 105°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

2296

QUESTION ID

S57-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 55°

GRADING NOTE (160 − 50)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

2296

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the formula for two tangents is a special case of the secant formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

2297

QUESTION ID
S57-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The tangent case is the limit where each secant's two intersection points merge; the arcs are the major and minor arcs.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

SET 58 • HSG-C.A.2 • CIRCLES

Relationships among radii and chords

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Right triangles: Hypotenuse-Leg and the Pythagorean Theorem	**HSG-SRT.B.5**
BACKWARD 2	The perpendicular bisector theorem	**HSG-CO.C.9**
BACKWARD 3	Radii of a circle are equal	**HSG-CO.A.1**
FORWARD 1	Locate the center of a circle from chords	**HSG-C.A.2**
FORWARD 2	Find chord lengths and central angles with trigonometry	**HSG-SRT.C.8**

SET 58 • HSG-C.A.2

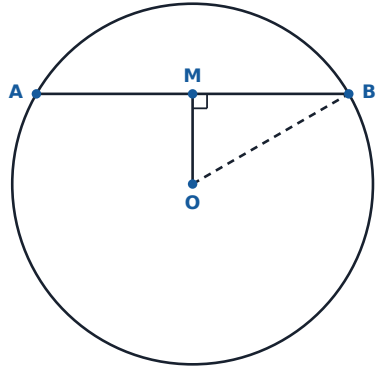
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, OM ⊥ chord AB at M. Prove that M is the midpoint of AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

2300

QUESTION ID

S58-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AM = MB

GRADING NOTE REQUIRED WORK: Prove △OMA ≅ △OMB (HL, using OA = OB).

Any valid proof (paragraph, two-column, or flow). Essential: OA = OB (radii); OM shared; right angles at M; HL (or the Pythagorean Theorem: AM = √(r² − OM²) = MB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

A circle has radius 13. A chord is 5 units from the center. How long is the chord?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

2301

QUESTION ID
S58-M-Q2

TYPE
Short answer

CORRECT ANSWER
24

GRADING NOTE Any equivalent form. Half-chord $\sqrt{(169 - 25)} = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

A chord of length 16 is in a circle of radius 10. How far is the chord from the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

2302

QUESTION ID

S58-M-Q3

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, chords AB and CD are congruent. OM ⊥ AB at M and ON ⊥ CD at N. Prove that OM = ON (congruent chords are equidistant from the center).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

2303

QUESTION ID

S58-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

OM = ON

GRADING NOTE REQUIRED WORK: Use the perpendicular-bisects-chord result, then HL on △OMA and △ONC.

Any valid proof (paragraph, two-column, or flow). Essential: OM ⊥ AB bisects AB and ON ⊥ CD bisects CD, so AM = ½AB = ½CD = CN; OA = OC (radii); right angles at M and N; HL gives △OMA ≅ △ONC, so OM = ON.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle O has radius 17. Chords AB and CD are each 30 units long. Find the distance from O to each chord, and explain why the distances are equal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

2304

QUESTION ID

S58-M-Q5

TYPE

Short answer

CORRECT ANSWER

Each distance is 8 (half-chord 15, √(289 − 225) = 8). They are equal because congruent chords are equidistant from the center.

GRADING NOTE Form required: the distance with work and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 1

Right triangles: Hypotenuse-Leg and the Pythagorean Theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangles with equal hypotenuses and one pair of equal legs are congruent by

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

2306

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

HL

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 10, leg 6. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

2307

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-B1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

True or false?

In circle O, OM is perpendicular to chord AB at M. The radii OA and OB are congruent hypotenuses of right triangles OMA and OMB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

2308

QUESTION ID

S58-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE OA = OB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

OM ⊥ AB in circle O. △OMA and △OMB share

A. nothing

B. OM

C. AB

D. OA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

2309

QUESTION ID

S58-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. OM

GRADING NOTE Common leg.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

True or false?

HL works for any triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

2310

QUESTION ID

S58-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 2

The perpendicular bisector theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

2312

QUESTION ID

S58-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

The center O of a circle is equidistant from the endpoints of chord AB. Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

2313

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

OA and OB are radii.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

The center of a circle is equidistant from the endpoints of each of its chords. So the center lies on

A. the perpendicular bisector of every chord

B. a tangent

C. the circle

D. every chord

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

2314

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the perpendicular bisector of every chord

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

Two chords of a circle are not parallel, so their perpendicular bisectors are different lines; the bisectors meet at P. What is P?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

2315

QUESTION ID

S58-B2-Q4

TYPE

Short answer

CORRECT ANSWER

The center of the circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of a chord can miss the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

2316

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It always passes through it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 3

Radii of a circle are equal • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center O. OA = 7. Find OB if B is on the circle.

OB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

2318

QUESTION ID

S58-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

True or false?

All radii of one circle have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

2319

QUESTION ID

S58-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

AB is a chord of circle O, and OA = 5. OB =

A. unknown

B. 10

C. 2.5

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

2320

QUESTION ID

S58-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

GRADING NOTE Radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

What is the longest chord in a circle of radius 6?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

2321

QUESTION ID

S58-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A diameter, 12

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

True or false?

A chord always passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

2322

QUESTION ID

S58-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

SET 58 • HSG-C.A.2

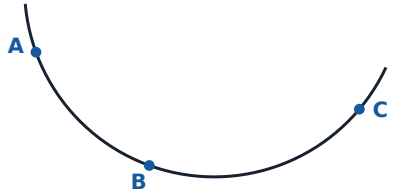
FORWARD 1

Locate the center of a circle from chords • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

The arc below is part of a circle. Construct its center.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

2324

QUESTION ID

S58-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw chords AB and BC; construct their perpendicular bisectors; they meet at the center.

GRADING NOTE Grade the construction marks, not only the final figure: two chords and both perpendicular bisector constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle passes through (0, 0), (8, 0), and (0, 6). Find its center.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

2325

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 3)

GRADING NOTE Form required: an ordered pair. Perpendicular bisectors x = 4 and y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

True or false?

Three points not on one line determine exactly one circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

2326

QUESTION ID

S58-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The perpendicular bisectors meet in one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

Why can't three points on one line lie on a circle?

A. Three points on a line always lie on a circle

B. Their perpendicular bisectors are parallel, so no center exists

C. Only two points can lie on a circle

D. They are too close

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

2327

QUESTION ID

S58-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Their perpendicular bisectors are parallel, so no center exists

GRADING NOTE Parallel bisectors never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A circular plate is broken. Describe how to find its original radius from the piece.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

2328

QUESTION ID
S58-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Mark three points on the edge, construct the perpendicular bisectors of two chords to find the center, then measure to the edge.

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSG-C.A.2

FORWARD 2

Find chord lengths and central angles with trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A chord subtends a central angle of 80° in a circle of radius 10. Find its length to the nearest tenth.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

2330

QUESTION ID
S58-F2-Q1

TYPE
Short answer

CORRECT ANSWER
12.9

GRADING NOTE Rounding to 0.1 required. Half the chord = 10 sin 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A chord of length 12 is in a circle of radius 10. Find the central angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

2331

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

73.7°

GRADING NOTE Rounding to 0.1° required. Half-angle $\sin^{-1}(6/10)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

True or false?

A chord's length is 2r sin θ for central angle θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

2332

QUESTION ID

S58-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 2r sin(θ/2), from the right triangle with half the chord.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Central angle 60°, radius 9. Chord length?

A. 4.5

B. 18

C. 9

D. $9\sqrt{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

2333

QUESTION ID
S58-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 9

GRADING NOTE $2 \cdot 9 \cdot \sin 30° = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

2333

ANSWER KEY Reduced copy of the question page, with its answer

A chord equals the radius. Find its central angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

2334

QUESTION ID

S58-F2-Q5

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form. Equilateral triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

SET 59 • HSG-C.A.3 • CIRCLES

Construct the inscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct an angle bisector	**HSG-CO.D.12**
BACKWARD 2	Construct a perpendicular from a point to a line	**HSG-CO.D.12**
BACKWARD 3	Points on an angle bisector are equidistant from the sides	**HSG-SRT.B.5**
FORWARD 1	Prove that the incenter is equidistant from the sides	**HSG-C.A.3**
FORWARD 2	Locate a point equidistant from three lines in a design	**HSG-MG.A.3**

SET 59 • HSG-C.A.3

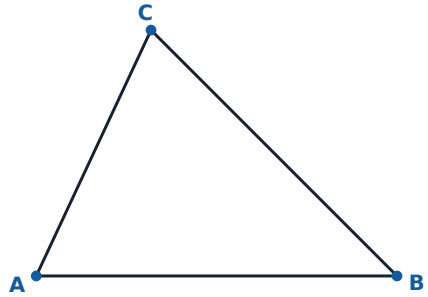
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

2337

QUESTION ID

S59-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect two angles (for example ∠A and ∠B); they meet at the incenter I; construct the perpendicular from I to one side, meeting it at D; draw the circle with center I and radius ID.

GRADING NOTE Grade the construction marks, not only the final figure: two angle bisector constructions, the perpendicular from I to a side, and the circle touching all three sides.

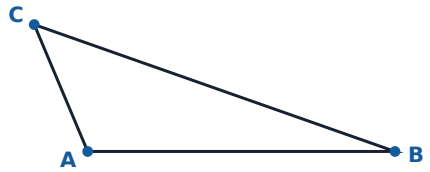
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of obtuse triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

2338

QUESTION ID

S59-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps: two angle bisectors meet at I (inside the triangle), perpendicular from I to a side gives the radius, then the circle.

GRADING NOTE Grade the construction marks, not only the final figure: all constructions and the circle.

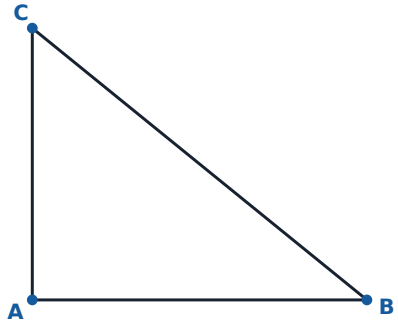
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of right triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

2339

QUESTION ID

S59-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Two angle bisectors, the perpendicular from their intersection to a side, and the circle.

GRADING NOTE Grade the construction marks, not only the final figure: all constructions and the circle.

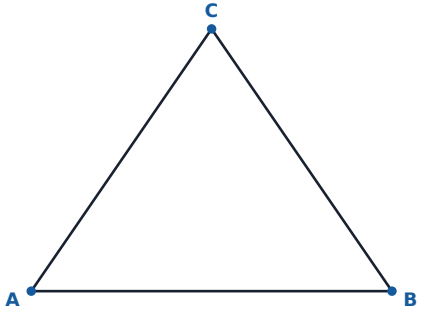
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of isosceles triangle ABC (AC = BC). Then, as evidence, construct the perpendiculars from the incenter to ALL THREE sides and mark the three points where the circle touches the sides.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5

2340

QUESTION ID

S59-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect two angles to get I; construct perpendiculars from I to AB, BC and CA (feet D, E, F); draw the circle with center I through D; it passes through E and F and touches each side once.

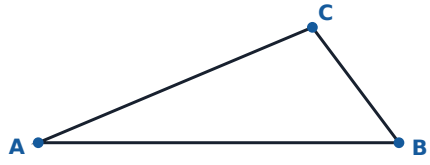
GRADING NOTE Grade the construction marks, not only the final figure: two angle bisectors, all three perpendiculars, and a circle through the three feet (evidence that the distances are equal).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of triangle ABC by bisecting ∠B and ∠C (not ∠A). Then construct the bisector of ∠A as a check.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

2341

QUESTION ID

S59-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

The bisectors of ∠B and ∠C meet at I; the perpendicular from I to a side gives the radius; the circle is drawn; the bisector of ∠A also passes through I (the check).

GRADING NOTE Grade the construction marks, not only the final figure: three angle bisectors meeting at one point, the perpendicular, and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 1

Construct an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

How many arcs are needed to bisect an angle with compass and straightedge?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

2343

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Three (one at the vertex, then two equal arcs from the crossing points)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

True or false?

An angle bisector construction ends with a ray from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

2344

QUESTION ID

S59-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Through the crossing of the equal arcs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The incenter is where the

A. medians meet

B. perpendicular bisectors meet

C. angle bisectors meet

D. altitudes meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

2345

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. angle bisectors meet

GRADING NOTE Equidistant from sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

Why construct only two angle bisectors to find the incenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

2346

QUESTION ID

S59-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The third bisector passes through the same point.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

True or false?

Bisecting the sides locates the incenter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q5/5

2347

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That locates the circumcenter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 2

Construct a perpendicular from a point to a line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction finds the point where the incircle touches a side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

2349

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular from the incenter to the side

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

True or false?

The perpendicular from a point to a line is the shortest segment to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

2350

QUESTION ID

S59-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

From point I, an arc meets side AB at two points. Next step?

A. Measure AB

B. Draw a circle at A

C. Bisect ∠I

D. Arcs of equal radius from those two points, crossing on the other side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

2351

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Arcs of equal radius from those two points, crossing on the other side

GRADING NOTE Then draw the line through I and that crossing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

Why not measure from I to a vertex for the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

2352

QUESTION ID

S59-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Vertices are farther than the sides; the circle must touch the sides, not pass through vertices.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

True or false?

The incircle passes through the vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

2353

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It touches the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 3

Points on an angle bisector are equidistant from the sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

True or false?

A point on the bisector of ∠A is equidistant from the two sides of ∠A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

2355

QUESTION ID

S59-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent right triangles (AAS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The incenter is 3 cm from side AB. How far is it from side BC?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

2356

QUESTION ID
S59-B3-Q2

TYPE
Short answer

CORRECT ANSWER
3 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

Distance from a point to a side is measured

A. to the nearest vertex

B. along the perpendicular

C. along the bisector

D. along any segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

2357

QUESTION ID

S59-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. along the perpendicular

GRADING NOTE Shortest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

Why is the incenter the same distance from all three sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

2358

QUESTION ID

S59-B3-Q4

TYPE

Short answer

CORRECT ANSWER

It is on all three angle bisectors, and each bisector point is equidistant from that angle's sides.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

True or false?

The incenter is equidistant from the three vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q5/5

2359

QUESTION ID
S59-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE That is the circumcenter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSG-C.A.3

FORWARD 1

Prove that the incenter is equidistant from the sides • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

The bisectors of ∠A and ∠B of △ABC meet at I. Prove that I is equidistant from all three sides and lies on the bisector of ∠C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

2361

QUESTION ID

S59-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

I is equidistant from AB, BC, CA and lies on the bisector of ∠C.

GRADING NOTE REQUIRED WORK: Use the angle bisector distance property twice, then its converse at C.

Any valid proof (paragraph, two-column, or flow). Essential: on the bisector of ∠A: d(I, AB) = d(I, AC); on the bisector of ∠B: d(I, AB) = d(I, BC); so d(I, AC) = d(I, BC); the converse puts I on the bisector of ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

A triangle has area 30 and perimeter 30. Find the radius of its inscribed circle.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5

2362

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Area = r · (perimeter ÷ 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5

True or false?

The incenter of an obtuse triangle lies outside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q3/5

2363

QUESTION ID

S59-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The incenter is always inside; the circumcenter is the one that can be outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q3/5

A 3-4-5 right triangle has inradius

A. 2.5

B. 1

C. 1.5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

2364

QUESTION ID

S59-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1

GRADING NOTE Area 6 = r · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why area = r · s, where s is half the perimeter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

2365

QUESTION ID

S59-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Joining I to the vertices splits the triangle into three triangles with heights r on bases a, b, c: area = ½r(a + b + c) = r · s.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSG-C.A.3

FORWARD 2

Locate a point equidistant from three lines in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A circular fountain must touch three straight paths that form a triangle with sides 13, 14, and 15 m (area 84 m²). What is the largest possible radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

2367

QUESTION ID

S59-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 m

GRADING NOTE Any equivalent form. s = 21, r = 84/21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

Explain why the incircle gives the largest circle that fits inside a triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

2368

QUESTION ID
S59-F2-Q2

TYPE
Short answer

CORRECT ANSWER
A larger circle would cross some side; the incircle touches all three sides.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

True or false?

A sprinkler equidistant from three straight fences forming a triangle should be at the circumcenter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

2369

QUESTION ID

S59-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the incenter (equidistant from lines); the circumcenter is equidistant from the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

To be equidistant from three towns (points), use the

A. orthocenter

B. centroid

C. incenter

D. circumcenter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

2370

QUESTION ID

S59-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. circumcenter

GRADING NOTE Points, not lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An equilateral triangle has side 6. Find its inradius exactly.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

2371

QUESTION ID
S59-F2-Q5

TYPE
Short answer

CORRECT ANSWER
√3

GRADING NOTE Form required: exact. Area 9√3, s = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

SET 60 • HSG-C.A.3 • CIRCLES

Construct the circumscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 2	Points on the perpendicular bisector are equidistant from the end…	**HSG-CO.C.9**
BACKWARD 3	Triangle types: acute, right, obtuse	**4.G.A.2**
FORWARD 1	Find the circumcenter and radius with coordinates	**HSG-GPE.B.5**
FORWARD 2	Find the circumradius with trigonometry (extended Law of Sines)	**HSG-SRT.D.10**

SET 60 • HSG-C.A.3

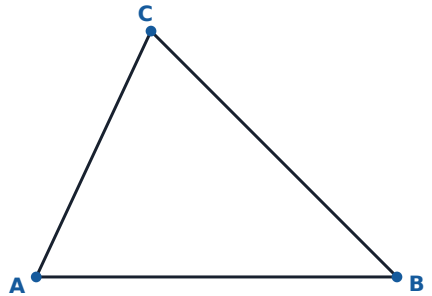
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

2374

QUESTION ID

S60-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular bisectors of two sides; they meet at O; draw the circle with center O through A (it also passes through B and C).

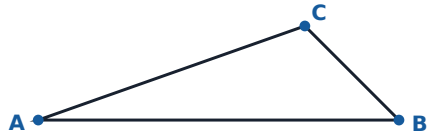
GRADING NOTE Grade the construction marks, not only the final figure: two perpendicular bisector constructions and the circle through all three vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of obtuse triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

2375

QUESTION ID

S60-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps; the circumcenter is outside the triangle.

GRADING NOTE Grade the construction marks, not only the final figure: all constructions and the circle.

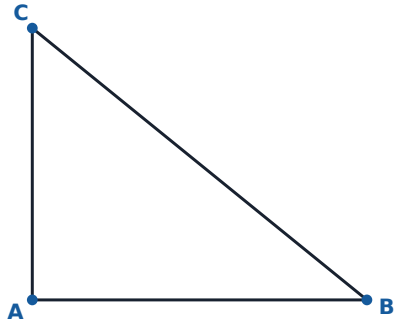
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of right triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

2376

QUESTION ID

S60-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The perpendicular bisectors meet at the midpoint of hypotenuse BC; circle centered there through A.

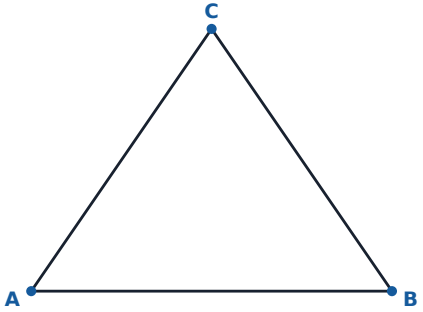
GRADING NOTE Grade the construction marks, not only the final figure: constructions and the circle; the center is on BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of isosceles triangle ABC (AC = BC). Then, as evidence, construct the perpendicular bisector of the third side and show that it passes through the same center.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

2377

QUESTION ID

S60-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Perpendicular bisectors of two sides meet at O; the third perpendicular bisector also passes through O; the circle with center O through A passes through B and C.

GRADING NOTE Grade the construction marks, not only the final figure: three perpendicular bisectors meeting at O and the circle through all three vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of triangle ABC using the perpendicular bisectors of AC and BC. Then check with the compass that the circle passes through all three vertices.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

2378

QUESTION ID

S60-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

The perpendicular bisectors of AC and BC meet at O, outside the triangle below AB; the circle with center O through C also passes through A and B.

GRADING NOTE Grade the construction marks, not only the final figure: both perpendicular bisector constructions and the circle through A, B and C; the center is outside the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 1

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Which construction gives all points equidistant from A and B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

2380

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector construction uses equal arcs from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

2381

QUESTION ID

S60-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The circumcenter is where the

A. angle bisectors meet

B. sides meet

C. perpendicular bisectors of the sides meet

D. medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

2382

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. perpendicular bisectors of the sides meet

GRADING NOTE Equidistant from vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

After constructing two perpendicular bisectors, why is the third unnecessary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

2383

QUESTION ID

S60-B1-Q4

TYPE

Short answer

CORRECT ANSWER

It passes through the same point.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

True or false?

The arcs in a perpendicular bisector construction can have any two different radii.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

2384

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 2

Points on the perpendicular bisector are equidistant from the endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

2386

QUESTION ID

S60-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

O is on the perpendicular bisectors of AB and BC.
Compare OA, OB, OC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

2387

QUESTION ID
S60-B2-Q2

TYPE
Short answer

CORRECT ANSWER
OA = OB = OC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The circumcircle's radius is

A. the distance from O to any vertex

B. the longest side

C. the distance from O to a side

D. half a side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

2388

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the distance from O to any vertex

GRADING NOTE Through the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

A circle passes through A, B, C. Where must its center lie relative to AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

2389

QUESTION ID

S60-B2-Q4

TYPE

Short answer

CORRECT ANSWER

On the perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The circumcenter is equidistant from the sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

2390

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE From the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 3

Triangle types: acute, right, obtuse • Grade 4 • Geometry • 4.G.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles 30°, 45°, 105°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

2392

QUESTION ID

S60-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Obtuse

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

True or false?

A right triangle has one 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

2393

QUESTION ID

S60-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

Where is the circumcenter of a right triangle?

A. Inside

B. Outside

C. On the hypotenuse (its midpoint)

D. At the right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

2394

QUESTION ID

S60-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. On the hypotenuse (its midpoint)

GRADING NOTE Hypotenuse is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles 50°, 60°, 70°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

2395

QUESTION ID

S60-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Acute

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

True or false?

An acute triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

2396

QUESTION ID

S60-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE All angles are acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 1

Find the circumcenter and radius with coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Triangle with A(0, 0), B(6, 0), C(2, 4). Find the circumcenter and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

2398

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 1); radius √10

GRADING NOTE Form required: both. Bisector of AB: x = 3; of AC: y − 2 = −½(x − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

Triangle with A(−2, 0), B(2, 0), C(0, 2). Find the circumcenter.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5

2399

QUESTION ID

S60-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 0)

GRADING NOTE Form required: an ordered pair. All three are 2 from the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5

True or false?

The circumcenter is the intersection of two perpendicular bisector equations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S60-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q3/5

2400

QUESTION ID
S60-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Solve the system.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S60-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q3/5